

MASTERARBEIT | MASTER'S THESIS

Titel | Title

Expert Persona Prompting: An Empirical Study of the Effects of
Expert Persona Prompt Design on Task-Oriented Performance

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angestrebter akademischer Grad | in partial fulfilment of the requirements for the degree of
Master of Science (MSc)

Wien | Vienna, 2026

Studienkennzahl lt. Studienblatt | Degree
programme code as it appears on the
student record sheet:

UA 066 921

Studienrichtung lt. Studienblatt | Degree
programme as it appears on the student
record sheet:

Masterstudium Informatik

Betreut von | Supervisor:

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Acknowledgements

First and foremost, I would like to thank my friends and give a special mention to B. Without their encouragement, this work would not have been possible.

I would also like to thank my parents. Their support over the past years made it possible for me to pursue my studies in the first place.

Finally, my gratitude also extends to my mentor, Pedro, who accompanied me over the last few months, offered valuable advice, and introduced me to scientific work.

Thank you!

Abstract

This thesis investigates how prompt design decisions affect model performance in expert-persona prompting. In particular, it examines three persona design choices: the length of persona strings, the audience level of audience-targeting personas, and the distinction between instance-based and domain-based personas. To address these questions, model performance is evaluated on five benchmarks across eight models from four model families. The analysis combines statistical evaluation and hypothesis testing using standard benchmark metrics with a qualitative examination of persona strings and model responses, enabling the thesis to address questions that cannot be answered through statistical methods alone. The results show that adding more detail to persona strings can improve task performance, that audience level has a significant impact on performance, and that instance-based personas generally outperform domain-based ones. However, the analysis also indicates that persona effects are model- and dataset-sensitive. The findings therefore suggest that persona prompt engineering can meaningfully alter model performance and responses, and that its effectiveness depends on several factors, including the dataset, the motivation for using personas, and available computational resources.

Kurzfassung

Die vorliegende Arbeit untersucht den Einfluss von Entscheidungen beim Prompt-Design auf die Modellleistung im Expert-Persona-Prompting. Dabei werden drei Entscheidungen des Persona-Designs untersucht: die Länge der Persona-Strings, das Zielgruppenniveau von zielgruppenorientierten Personas sowie die Unterscheidung zwischen instanzbasierten und domänenbasierten Personas. Zur Beantwortung dieser Fragen wird die Modellleistung von acht Modellen aus vier Modellfamilien anhand von fünf Benchmarks evaluiert. Die Analyse verbindet statistische Auswertungen und Hypothesentests auf Basis standardisierter Benchmark-Metriken mit einer qualitativen Untersuchung von Persona-Strings und Modellantworten. Dadurch lassen sich auch Fragen behandeln, die sich nicht allein mit statistischen Methoden beantworten. Die Ergebnisse zeigen, dass ein höherer Detailgrad in Persona-Strings die Modellleistung verbessern kann, dass das Zielgruppenniveau einen signifikanten Einfluss auf die Leistung hat und dass instanzbasierte Personas im Allgemeinen besser abschneiden als domänenbasierte. Die Analyse zeigt jedoch auch, dass Persona-Effekte modell- und datensatzabhängig sind. Die Ergebnisse legen daher nahe, dass Persona-Prompt-Engineering Modellleistung und Modellantworten wesentlich verändern kann, zugleich aber seine Wirksamkeit von mehreren Faktoren abhängt, wie zum Beispiel dem Datensatz, der Motivation hinter der Verwendung von Personas und den verfügbaren Rechenressourcen.

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1. Motivation

Personas are a simple concept: descriptions of roles that guide a large language model (LLM) in how it should behave and formulate its answers, similar to stage directions for an actress in a performance. Despite their simplicity, they open a fascinating field of study. For instance, personas allow LLMs to exhibit human-like traits (Dash et al., 2025) or engage in role-play with consistent personalities guided by their own principles and beliefs (Gupta et al., 2024; Wang et al., 2024a). These characteristics enable many useful applications, such as examining stereotypes LLMs associate with the role a persona should convey (Gupta et al., 2024; Kamruzzaman et al., 2026). These bias studies are an especially critical field, as developing strategies for bias mitigation is an important step toward creating safe LLMs (Ouyang et al., 2022; Vijjini et al., 2025).

In another direction, personas have shown benefits in enabling LLMs to reproduce the knowledge and expertise of real-world professionals. Several studies associate the use of so-called *expert personas* with improved performance in question–answering tasks (Kong et al., 2024; Xu et al., 2023). However, it remains unclear to what extent persona prompting truly helps and which conditions prove beneficial in improving model performance. One study suggests that their usefulness is limited to certain knowledge domains, such as politics or biology (Kong et al., 2026). Furthermore, when comparing the work of two research groups, it is evident that outcomes vary even depending on the model used in the experiments. The studies in question report directly opposing results (Han and Wang, 2024; Kong et al., 2024).

Nevertheless, a baseline consensus reached in several cases is that personas generally lead to increased performance. The main challenge lies in identifying the correct role and determining which other factors influence the outcome (Luz de Araujo et al., 2025; Zheng et al., 2024). With this in mind, this thesis seeks to explore how prompt design influences whether expert personas are useful by examining several design aspects.

The first aspect in question is the persona string length, that is, to what extent the length or detail of the persona description affects the capabilities of an LLM. A previous publication found a positive correlation between the length of the persona description and the performance gain. The study, however, restricts experiments to mathematical reasoning benchmarks, and mentions the persona length only briefly (Kong et al., 2024). As such, further experiments could provide additional insights into exactly how persona length and model performance correlate.

The aforementioned study also employed *teacher personas*. In this concept, the role shifts from simply being an expert to being a figure who explains concepts, ideas, or answers questions. Interestingly, such teacher personas, paired with an expert identity, have been shown to perform better than the expert identity alone. Building on this foundation, Zheng et al. (2024) conducted a study comparing audience- and speaker-

1. Motivation

specific roles. Their surprising finding was that instructing the model to *speak* to someone performed better than telling it to *be* someone. This thesis extends this observation by examining whether varying this audience level changes the conclusion.

As a third and final persona design choice, the thesis uses personas of varying lengths in both domain-specific and instance-specific forms. The standard terminology refers to these two persona types as static personas, which are domain-specific, and dynamic personas, which are instance-specific. Although previous research indicates that dynamic personas reach higher performance scores than static personas (Luz de Araujo et al., 2025), this distinction remains relevant for further testing for two reasons. Firstly, the analysis of persona length effects is conducted separately for static and dynamic personas. Secondly, examining both persona types allows testing whether the previously mentioned performance differences also hold under the specific conditions used for the experiments here.

In summary, this thesis explores three main research questions:

1. How does the length of an expert persona string and LLM performance correlate on open and closed tasks?
2. To what extent does the audience expertise level of a teacher persona affect task performance?
3. What influence do static and dynamic personas have on model performance?

This thesis uses a total of eight different models from four model families to carry out the experiments, some of which are open-weight models. To obtain a broad perspective on the effects of these traits, the benchmarks include open- and closed-format tasks. The first such benchmark is **MMLU-Pro** (Wang et al., 2024b), which consists of closed-format single-choice questions from domains such as business, health, or history, with a varying number of answer options. The evaluation also includes the more specialized **MATH** (Hendrycks et al., 2021b) dataset, which evaluates mathematical reasoning and open-ended problem-solving. To further assess instruction following capabilities, **IFBench** (Pyatkin et al., 2025) and **AlpacaEval** (Li et al., 2023) are used as well. Finally, the **FLORES+** (Team et al., 2022) dataset tests the machine-translation abilities of the personas.

To provide quantitative insights into the research questions, standard performance metrics, such as accuracy and win rate, are used. The latter case involves LLM-based judges to assess the performance of personas. For this, a cross-model format ensures the judgments remain unbiased, in which the models that produce the answers differ from the model that evaluates them. Prior work shows that judges are biased toward selecting answers from their own model family (Panickssery et al., 2024). To further mitigate bias, the judgment prompts contain shuffled completions.

To summarize, the main contributions of this master’s thesis are:

- A systematic evaluation of expert persona design decisions across eight LLMs from four model families, including both open- and closed-weight models.

- An analysis of how the persona design choices, namely persona length, teacher persona audience level, and static versus dynamic persona generation, affect model performance across different tasks.
- A combined quantitative and qualitative discussion of persona prompting effects, aiming to explain not only whether certain personas improve performance, but also under which conditions and why these effects may occur.

The remainder of the thesis has the following structure. Chapter 2 presents related work. It begins with a general overview of persona prompting by introducing a loose taxonomy of persona types based on their use cases. A discussion of persona generation and application methods follows, before going over previous findings on the influence of persona prompting on model performance. Chapter 3 introduces the problem in an abstract form and presents the concrete persona types used throughout the experiments. It further describes the experimental setup, including the selection of benchmarks and models, the persona generation procedure, the evaluation metrics, and the statistical methods used to answer the research questions. Chapter 4 then reports the experimental results, including performance measurements and statistical test outcomes. Chapter 5 further discusses these findings, where they are also related to prior work. The thesis closes with a conclusion in Chapter 6.

2. Related Work

Before going deeper into the experimental setup and the considerations that guide it, it is necessary first to introduce the central concept of this thesis: personas in the context of large language models. What are they?

In general, personas are textual descriptions that instruct a model to adopt a particular role, based on some identity or set of characteristics, when generating responses. These descriptions may include information about expertise, personality traits, communication style, or social roles, with the goal of shaping both the content and the form of model outputs.

To further answer the above question, this chapter provides an extensive overview of the research progress on persona prompting. Each following section focuses on one aspect of persona usage in LLMs, beginning with Section 2.1, which introduces a taxonomy of persona types based on their use cases, anticipated benefits, and implications, while illustrating the different forms personas can take. It clarifies what types of personas exist and the motivations behind their use.

This brief introduction continues with Section 2.2. Questions about the generation and application of personas, addressing how LLMs are supposed to elicit any positive or negative effects, guide this part of the thesis. One such example is persona prompting, the fundamental prompting strategy, where the user instructs a model to assume a role similar to instructing it to solve a task. The section discusses a range of additional examples.

Section 2.3 reviews empirical findings on the effects of persona prompting on model performance, covering both improvements and drawbacks. This analysis evaluates whether the anticipated benefits are actually achieved, and further motivates the research questions by contextualizing the works of previous authors, thereby identifying a gap in the current research on prompt design.

Finally, Section 2.4 introduces explanation-based prompting in conjunction with so-called teacher personas, establishing the foundation for the thesis’s research question on audience-based personas.

2.1. Persona Types and Application Scenarios

When models were first able to participate in multi-turn dialogues, researchers quickly observed their limited ability to maintain stable personalities (Vinyals and Le, 2015). Although this was a few years before researchers began examining personas, it foreshadowed later developments, as early work on persona prompting focused primarily on addressing this limitation (Welleck et al., 2019; Song et al., 2020; Zhou et al., 2023b). These efforts proved to be worthwhile, as they not only improved consistency but also had a positive

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impact on the relevance and informativeness of model responses (Song et al., 2020; Zhou et al., 2023b). An intuitive outcome, as models that fail to remain in character are more likely to shift topics or list irrelevant information.

However, improvements were not limited to the semantic properties; the lexical characteristics were also affected. By conditioning models on personas, researchers observed increased diversity in generated responses, which contributed to more human-like dialogues (Song et al., 2019). Such developments marked an important first step toward creating more engaging conversations, as such diversification aligns more with human conversational preferences (Lu et al., 2022). From this initial phase of research, several other application scenarios emerged.

2.1.1. Expert Personas

The first such application scenario is the use of expert personas, which are roles that are related to some domain by the level of expertise they have in it. Several studies have shown both positive and negative effects of these expert personas on model performance (Kim et al., 2025; Luz de Araujo and Roth, 2025; Kong et al., 2024, 2026; Wang et al., 2024c; Xu et al., 2023). The motivation behind this type of persona is to test whether models can better elicit expert knowledge when prompted from such a perspective.

These personas are often composed of an expert identity, such as some specific occupation (Wang et al., 2024c; Xu et al., 2023) or a scholar in an area of research (Xu et al., 2023). More broadly, the term expert should not be limited to theoretical knowledge: Another example uses explicit expert personas for the different roles of a team that collaborates on a more complex task (Qian et al., 2024). In more advanced cases, persona strings include textual descriptions of expertise, domain knowledge, and the skills required to solve a particular question (Kong et al., 2026; Xu et al., 2023), as well as personality traits associated with certain occupations (Kang et al., 2024). Section 2.1.3 presents further examples. This section highlights how the distinction between persona types can become blurred in some cases. Specifically, personas can represent hobby experts that are knowledgeable in a fandom (Wang et al., 2024c), which is useful for role-play. Pairing this type of persona with instructions on how to answer a question (Kong et al., 2024) or other prompting strategies (Han and Wang, 2024) can further enhance positive effects.

Their primary application spans many different forms of task-solving, ranging from summarization (Xu et al., 2023) to reasoning tasks or general instruction following (Kim et al., 2025; Luz de Araujo and Roth, 2025; Kong et al., 2024, 2026). They are also used to improve performance in creative tasks, with studies showing satisfying outcomes in creative writing (Wang et al., 2024c), software engineering (Qian et al., 2024), and even portfolio management (Abe et al., 2024).

2.1.2. Personas Based on Demographical Traits

A textual description of a role can carry a lot more information than some plain occupation. Such descriptions can define an identity, which a human might feel connected to, for example, some religion or gender. In the real world, such identities often face discrimination

2.1. *Persona Types and Application Scenarios*

and bias. Since the training data of LLMs mostly comes from unmoderated online resources, a natural next question is whether LLMs show biases against these demographics.

For these types of bias studies, personas are usually based on demographic traits, such as political affiliation, age, ethnicity, religion, disabilities, ideology, social classes, sexuality, gender, and many others (Luz de Araujo and Roth, 2025; Deshpande et al., 2023; Gupta et al., 2024; Kamruzzaman et al., 2026; Salewski et al., 2023; Zheng et al., 2024). In other settings, they might also be roles defined by some relationship, such as mother or mentor (Zheng et al., 2024), or simply names in a certain language (Luz de Araujo and Roth, 2025). While these examples do not carry any demographic traits explicitly, they are still there. A name may still implicitly signal a particular ethnicity. They can also be character-based personas (Deshpande et al., 2023) or be defined by personality traits (Yeo et al., 2025).

The intention behind demographic personas is not necessarily to improve model performance, as is hoped for with expert personas. Instead, they are mainly used to analyze and investigate stereotypes models associate with these roles, for example, by examining how sensitive models are to misinformation (Haupt et al., 2024; Salewski et al., 2023), how personas influence model refusal behavior (Plaza-del Arco et al., 2025), or how a model’s responses on instruction tasks change when assigned a demographic persona (Luz de Araujo and Roth, 2025; Deshpande et al., 2023; Gupta et al., 2024; Haupt et al., 2024). It is important to understand how LLMs behave in this respect for fairness and safety reasons, as most of these biases are unwanted. As such, these studies form the basis for developing mitigation strategies (Kamruzzaman and Kim, 2025). Apart from this, they have also been used to simulate human behavior in, e.g., opinion surveys (Koncel-Kedziorski et al., 2025).

2.1.3. **Role-Playing Personas**

Building directly on the initial line of research, another application scenario is the use of personas in role-playing (Shanahan et al., 2023). The variants of this type are highly diverse: they may possess any imaginable character trait (Chan et al., 2024; Ge et al., 2025; Wang et al., 2024a; Zhou et al., 2023b), be certain living individuals (Deshpande et al., 2023), or fictitious characters (Wang et al., 2024a).

These methods extend earlier research primarily by aiming to enhance semantic and lexical diversity, which is useful, among others, in generating synthetic data (Ge et al., 2025) or more complex benchmarks with more human-like question formulations (Li et al., 2024). A more advanced setting for this type of persona is LLM personalization, which aims to provide a custom user experience (Cheng et al., 2025; Wang et al., 2024a). In the latter case, personas sometimes evolve based on previous human-machine interactions. As a result, they do not necessarily need to be statically assigned through persona prompting. Instead, they might also be inferred from user input (Chen et al., 2024; Cheng et al., 2025), either explicitly, by providing relevant user information, or implicitly, by allowing the model to infer from the context of prior interactions (Vijini et al., 2025).

2.2. Persona Construction and Application

Since the introduction of personas in LLM research, creating persona-consistent models has been a central research focus. It therefore does not come as a surprise that many different frameworks have been proposed, both for generating and for assigning personas. This section presents a few important examples.

The simplest approach to persona generation involves hand-written personas by a human annotator. These persona strings are generally short and broadly defined, for example covering an entire field of study in the case of expert personas. Persona prompting allows researchers to assign personas of this type to an LLM. In a typical setup, the prompt instructs the LLM to assume a role and provides a textual description of the persona either as a prefix to another instruction (Luz de Araujo et al., 2025) or in a two-turn dialogue (Kong et al., 2024).

Since the static definition of personas may not be optimal for the task at hand, dynamic examples have been proposed as well. These are created automatically by an LLM through instance-based conditioning (Luz de Araujo et al., 2025). In the simplest approach, they are generated through prompting, e.g., with a specific instruction to solve (Luz de Araujo et al., 2025) or a personality trait (Deshpande et al., 2023) in mind. Other examples rely on in-context learning to produce more nuanced persona descriptions (Xu et al., 2023).

More advanced approaches to persona generation and assignment also exist. In fact, some of the earliest studies on personas examined how to train models to identify and exhibit personality traits (Lu et al., 2022). This area of research remains active and has led to the development of many other methods: More sophisticated ones combine fine-tuning, in-context learning, and persona prompting (Wang et al., 2024a), or even directly influence model activations through matrix–vector calculations (Potertì et al., 2025).

Models are also expected to adopt personas autonomously. One such example utilizes a model fine-tuned on a custom dataset that enables the LLM to infer the persona from context and assume the role through a monologue (Kong et al., 2026).

Beyond simple persona generation and assignment, some methods explicitly aim to improve LLM performance. Jekyll & Hyde is a hybrid approach that combines the outputs of a domain-specific persona-assigned model and a no-persona model, with a third evaluator model deciding which answer is more accurate (Kim et al., 2025). A similar idea is used by Solo-Performance Prompting (Wang et al., 2024c) and ChatDev (Qian et al., 2024), with the distinction that a single LLM takes on multiple personas, simulating a multi-turn dialogue between them to identify information required for task-solving (Wang et al., 2024c).

2.3. Previous Research on Persona Prompting

In the past few years, an extensive portion of work has been put into examining the benefits and drawbacks of personas. In this section, several observations are presented and contextualized by comparing the findings of different authors. The form of study

depends on the application scenario, persona type, and task type.

2.3.1. Bias Studies

Although bias studies are not the primary focus of this thesis, they are a noteworthy area of research in persona prompting. In this form of analysis, we prompt the model to answer from the perspective of a certain role based on some of the demographic traits mentioned in Section 2.1.2. We essentially ask the model how it expects a demographic group would react to some prompt, which lets us systematically evaluate what types of stereotypes and biases LLMs associate with these groups. For example, we can test whether personas impact refusal rates or guardrails against malicious behavior.

Many different demographics seem to have a negative impact on model performance. Simply by telling the model to answer, for example, from the perspective of a transgender person, the accuracy starts to decline (Gupta et al., 2024). Such results are measured even though no ordinary developer would intentionally train these biases into models. They are intrinsically present only through the training data, which raises the question of where these biases originate.

In a minority of cases, these biases come from the expected knowledge level of the persona. For example, on reasoning tasks, researchers were able to show a monotonic increase in model performance as the persona’s age increased (Salewski et al., 2023). Another study found that personas holding higher diplomas were ranked more favorably (Zheng et al., 2024). Both of these findings seem intuitive, as humans with a higher education have a deeper understanding of their area of expertise. The same is true for the age progression of a human, with a few exceptions.

However, this explanation is insufficient, as measuring other forms of biases quickly leads to questionable outcomes. In one specific example, an LLM refrained from answering physics-related questions when being instructed to act as a religious person, stating that the correct answer conflicted with their belief (Gupta et al., 2024). Researchers made similar observations based on the political affiliation of the role, where it was shown that models are more prone to false conclusions if the correct one would oppose their ideology (Dash et al., 2025). While these responses may appear plausible to some extent, their consequences are significant.

Biases range from unexplainable to highly problematic. In one specific example, an LLM refrained from answering some questions, claiming an assigned disability hinders it from completing the task, even though the disability was irrelevant to the task (Gupta et al., 2024). This behavior is an extreme example of a bias that is present in many forms. In more mild cases, beauty standards seemed to influence the cultural knowledge exhibited by models (Kamruzzaman et al., 2026).

More broadly, personas have exposed many different biases. LLMs tend to show biases toward certain political and religious affiliations, ethnicities, or genders (Luz de Araujo and Roth, 2025; Gupta et al., 2024; Zheng et al., 2024). These effects can be arbitrary: for instance, when classifying misinformation, an atheistic persona performed better than a general religious one, but directly stating a particular denomination led to even better results (Haupt et al., 2024).

2. Related Work

Personas can affect toxicity levels, e.g., with male personas being more toxic than others (Deshpande et al., 2023). Personas further influence a model’s ability to refuse harmful requests (Luz de Araujo and Roth, 2025; Vijjini et al., 2025; Plaza-del Arco et al., 2025), which is particularly concerning since previous work focused on limiting such traits, and personas weaken these safety guardrails. Some roles even increased toxicity when they should not have, such as "a good person", though this effect was small. Looking at more malicious personas, e.g., "a bad person", the toxicity increased quite drastically (Deshpande et al., 2023).

These findings suggest that personas broadly influence biases. Answering the question of where they come from is a difficult task. A simpler answer can be found to the question of what influences biases the most: the model family. Ever since the first bias studies have been conducted, developers have started to limit them as much as possible. If we look at the outcome of one study, we can see how certain demographics used to show almost no biases on previous model versions. On newer ones, all of a sudden, these biases are present again (Plaza-del Arco et al., 2025).

Given these findings, an active community is working on mitigation strategies to reduce persona-induced biases. In addition to incorporating debiasing procedures into model training through advanced techniques, such as direct preference optimization (Vijjini et al., 2025), even simple prompting strategies can already lead to desirable outcomes (Kamruzzaman and Kim, 2025). For example, combining expert personas with other demographic traits reduced the biases to non-significant levels (Gupta et al., 2024). Other strategies have not proven to be beneficial, such as chain of thought (CoT), which shows no improvements (Dash et al., 2025).

Finally, it is important to note that personas do not necessarily worsen stereotypes or biases; some personas can even show a lower bias toward their own group compared to no persona at all (Luz de Araujo and Roth, 2025).

2.3.2. Personas in Task-Oriented LLM Evaluation

From a task-oriented perspective, research primarily focuses on expert personas with the intention of improving a model’s task-solving skill (Luz de Araujo et al., 2025). The influence of these expert personas can range from negative to non-significant to positive. The question is therefore not whether they have an impact, but what variables are most important to achieve higher benchmark scores.

For reasoning tasks, such as mathematical or commonsense reasoning, in-domain experts have been repeatedly associated with improved performance compared to other prompting strategies, such as CoT (Kong et al., 2024). In this context, several studies have examined how much the level of detail in the persona description or the relevance of the assigned role contributes to performance improvements, yielding intuitive results in many cases (Luz de Araujo et al., 2025; Kong et al., 2024; Zheng et al., 2024).

The second property, role relevance, proves to be particularly important if our goal is enhancing model performance. An irrelevant role can pose quite a challenge in this respect, as non-domain experts are mostly associated with a negative impact (Luz de Araujo et al., 2025; Zheng et al., 2024). But even domain-aligned personas do not necessarily outperform

2.3. Previous Research on Persona Prompting

a no-persona baseline (Luz de Araujo and Roth, 2025), which demonstrates that there is no universal explanation of how or when expert personas improve LLMs. Multiple-choice question domains further highlight this inconsistency: in some domains a clear advantage of expert personas is shown, in others there is no change to the accuracy at all or a performance decrease (Kong et al., 2026). In a very similar fashion, an in-domain role defined by personality traits rather than an expert occupation shows higher variability in performance scores with a negative to non-significant impact (Kang et al., 2024). These findings suggest that using an arbitrary persona without prior empirical evidence of their effects is not advisable and generally discouraged.

The level of detail and content of the persona description also remain a crucial consideration that has received some spotlight recently. As mentioned previously, a more nuanced persona description can lead to higher performance gains, at least according to one study, where the authors report the accuracy for several benchmarks and observed a monotonic increase with a respective increase in persona description length (Kong et al., 2024).

But the length is not the only variable that is relevant when examining the degree of detail, as other considerations can also shape the content: for example, in the context of expert personas, the level of specialization. A study exemplified this, finding a non-significant to positive impact on model performance, again with a larger increase in model performance with more niche experts (Luz de Araujo et al., 2025).

Whether an expert persona has a positive or negative impact is not only affected by the persona description. Another factor is the model itself. Two specific studies showcase this, both of which partially used the same benchmark datasets. The studies compare the accuracy of persona prompting on these datasets against other strategies, such as CoT, and report directly opposing results. The first study claims that zero-shot CoT is superior to personas, and combining CoT with role-playing is even worse than using persona prompting or CoT on its own (Han and Wang, 2024). The other finds that exactly the role-playing itself can be a useful CoT trigger, observing higher accuracies when using personas and concluding that this trigger leads to better performance (Kong et al., 2024). The only thing that sets the experiments apart is the models used. The study that claims persona prompting is less favorable uses significantly smaller models.

One property that appears relatively consistent is the generation method of personas, whether a persona is instance-based or domain-based. The superior technique appears to be instance-based personas due to two reasons. First, they tend to produce larger performance gains, which is the most desirable effect they can possess (Luz de Araujo et al., 2025). But even in cases where observing this outcome is not possible, they showed another useful property: their accuracy scores are less unstable (Kim et al., 2025). With a smaller variance, any positive or negative impact becomes more predictable, which shows that using these dynamic experts is a better choice and they are generally favorable to domain-based ones, also since this requires less manual labor.

Given these findings, ongoing research continues to investigate the effects of personas on LLMs, as they are a simple approach to increasing model performance, with many variables that shape the way they interact.

2. Related Work

2.3.3. Personas and Open Tasks

This thesis defines open tasks as tasks that do not have a clear distinction between correct and incorrect, or instructions that contain complex tasks without answer options. In this broad field of tasks, personas are particularly effective.

Considering the nature of human dialogue, this is not surprising. Each person’s personality directly influences how they communicate. LLMs, on the other hand, are trained on text corpora written by numerous human authors without any consistent character traits in mind. Personas therefore provide a way of approximating this diversity in expression, influencing lexical properties such as fluency, relevancy, and consistency (Lu et al., 2022; Song et al., 2020).

This enhancement enables their use on a wide range of creative tasks, such as writing fictional stories that require niche information. Personas can guide responses with the expertise associated with their roles, e.g., by being part of a fandom and thereby knowing information about fantasy characters that are part of the franchise. However, the effectiveness of personas appears to depend strongly on the underlying model. Only with larger or more capable LLMs does the creative writing ability seem to be enhanced (Wang et al., 2024c).

Nevertheless, many use cases arise from the previously mentioned lexical properties. They enable a variety of applications: personas positively impact the software engineering skills of an agent, leading to more executable code. The authors claim that personas even have the largest impact on the quality of the generated code (Qian et al., 2024). In other cases, researchers have successfully used personas for portfolio management (Abe et al., 2024) or imitating humans for LLM evaluation (Li et al., 2024).

This ability to imitate human behavior also allows more extensive studies of contexts where human-like dialogue is needed. It can be used for simulating environments like psychotherapy in a safe setting, with fewer ethical concerns, as no humans are harmed during the process, while also speeding up evaluation procedures (Steenstra et al., 2026).

For certain open instruction-following tasks, expert personas can also be beneficial. In one example, they produced more detailed summarizations, which led to a higher win rate when using LLM-as-a-judge (Xu et al., 2023). With specific personality traits, personas can additionally be used to let LLMs adopt certain stances in discussions (Chan et al., 2024).

These properties make personas valuable for LLM personalization, which is a more challenging task. User preferences vary widely, as the requirements are highly subjective. This adaptability requires the combination of many of the previously mentioned aspects, all of which personas can be beneficial to, forming the basis for customized LLM companions (Cheng et al., 2025; Wang et al., 2024a).

The previous examples mostly use personas in complex settings. While a clear trend of benefits is observable, it remains difficult to isolate which components of a persona description are responsible for these improvements. Domain expertise, however, appears to be a consistent factor in all cases. Beyond this, the contribution of other persona attributes is less well understood. The generation method, as with closed tasks, also appears to play its part: again, dynamically constructed personas yield better performance

(Wang et al., 2024c). This lack of knowledge highlights a gap in current research: a more fine-grained understanding of how individual persona attributes shape model behavior is still missing, motivating the analysis conducted in this thesis.

2.4. Self-Teaching LLMs & Teacher Personas

Explaining concepts, answering questions, or simply getting feedback on some work. All these tasks are undoubtedly part of what users ask LLMs to do. And all these tasks have a common denominator: they are the work of a teacher. Therefore, evaluating how models behave when being told to act like teachers is a logical consequence. But does this form of prompting have any impact on models, and if so, which ones? This section answers these questions.

A first indication that such explanation-based prompting can improve model performance comes from an analysis of the role relevance of personas, mentioned previously in Section 2.3.2. More relevant roles are associated with improved mathematical reasoning abilities; in addition, combining expert identities with teacher-like roles appears to enhance results further. While the exact mechanism remains unclear, this effect can be related to other prompting strategies such as chain-of-thought prompting. However, rather than generating reasoning steps for itself, explanation-based prompting frames the reasoning as being communicated to a listener (Kong et al., 2024).

Nevertheless, these observations remain hypothetical. A study on a proposed prompting strategy that explicitly requires the model to explain the task before providing the solution provides more direct evidence. In this approach, the authors introduce a prompt that first instructs the model to restate and analyze the question in its own words before solving it. The outcome is consistent improvements across a range of reasoning benchmarks. The above hypothesis seems confirmed in this particular case, though the effectiveness of this strategy depends on the length of the explanation: more extensive explanations are not the best choice in all experiments (Yugeswardeenoo et al., 2024).

Researchers observed a similar trend in broader knowledge-based evaluations, such as on the MMLU dataset. In this setting, the authors compare what difference there lies in making the model assume a role based on the audience it is talking to instead of being the role directly. The authors again observe a small but significant performance improvement (Zheng et al., 2024).

Beyond inference-time prompting, researchers have also explored the idea of models acting as teachers through explanations in training settings. In this context, models generate explanations that serve as an additional supervision signal, either for themselves or for other models. Incorporating such explanations during training leads to consistent performance improvements, particularly on reasoning-intensive tasks, which again highlights the usefulness of such explanations (Zhao et al., 2024).

From this progress, it becomes clear that explanation-based prompting and teacher-like personas can have a mostly positive effect on model performance. A research direction that remains underexplored is what influence the audience level has on model performance, which motivates the third research question.

2.5. Summary

Above all, the previous analysis of current research has shown one thing: the influence of any personas is sensitive to many different factors. These factors include internal ones, such as the persona role, as well as external ones, such as the model family used. Nevertheless, the thesis identified a few gaps in the current research.

The prompt design has been established as one of the most impactful aspects when trying to maximize positive benefits, which is proven through two examples: the role relevance and the level of specialization in the case of expert personas. Non-relevant roles have been associated mainly with a negative impact on performance (Luz de Araujo et al., 2025; Zheng et al., 2024), while a more specialized role seems to provide larger accuracy boosts (Luz de Araujo et al., 2025). The building block that is now missing is the description length. That is, whether the persona string length has any impact on the desired outcome. A first study has tried to analyze this aspect, but in a very limited manner (Kong et al., 2024). This gap motivates the first research question of this thesis.

Another promising direction is explanation-based prompting. In the previous section, the mostly positive trends of this prompting strategy were shown (Yugeswardeenoo et al., 2024; Zheng et al., 2024). If we combine this with the progress on expert personas from Section 2.3.2, which mentions the effects of persona specialization levels, another one of the central research questions follows. If increasing the level of specialization in expert personas leads to improved performance, does a similar effect hold for teacher personas when varying the level of the intended audience?

This thesis addresses these gaps by systematically investigating the variations in persona design, specifically the description length for expert personas and the audience level for teacher personas. In the following chapter, the experimental setup will further clarify the concrete details of how this thesis plans to achieve this goal.

3. Methodology

The following chapter presents all technical aspects guiding the experiments of this thesis. The main research questions of this thesis all focus on the prompt design of expert personas, which requires a thorough explanation of the types of personas evaluated. As such, the chapter starts with Section 3.1, which is particularly important. It covers information about what parts of the persona descriptions vary, which is the basis for evaluating the effects of the mentioned prompt design aspects.

The experimental setup is next introduced, first as a formal problem definition in Section 3.2. In Section 3.3, the practical perspective of the experiments is then explained. It contains a short presentation of models used, considerations that guided the choice of datasets, information about the persona generation process, and which dataset requires which metric. This section will also provide insights about how the models are able to leverage benefits from personas.

A portion of the experiments contain open-format tasks, where deciding whether a model generated a good answer goes beyond simply comparing it to the ground truth. For the relevant benchmarks, this thesis uses LLM-as-a-judge (Zheng et al., 2023), which is a type of assessment methodology where the responses of LLMs are ranked by other models. Section 3.4 will present a description of this strategy.

3.1. Persona Properties

The centerpiece of the experiments of this thesis are personas. As highlighted in the Related Work section, researchers have come up with a large variety of persona prompting tools. This includes strategies for generating personas, the role a persona reflects, more specifically the content of the textual description of a persona string, and ways of applying them to an LLM. Taking these considerations into account, the following overview explains the types of personas and persona properties used for experimentation. As a preliminary point, all personas fall into the broader category of expert personas.

3.1.1. Static and Dynamic Personas

The distinction between static and dynamic personas is made based on the scope of expertise of expert personas. A static persona is a persona description that is tailored to a whole domain of tasks. This thesis therefore defines static personas as personas that are constant for a whole collection of questions from this domain. In other words, given a set of questions from a single domain, the same static persona would be generated for all of them by conditioning the persona-generating LLM with the domain. Thus, there is a one-to-one correspondence between question domain and persona.

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Base	You are a legal advocate
Short	<i>You are a legal advocate who relentlessly analyzes statutes, interprets precedent, and advocates for clients with integrity, precision, and strategic judgment.</i>
Medium	<i>You are a legal advocate who relentlessly analyzes statutes, interprets precedent, and advocates for clients with integrity, precision, and strategic judgment. You anticipate opposing arguments, assess risk, and tailor strategy to maximize ethical advocacy and client outcomes. You communicate clearly with clients, courts, and colleagues, translating complex doctrine into persuasive, principled conclusions.</i>
Long	<i>You are a legal advocate who relentlessly analyzes statutes, interprets precedent, and advocates for clients with integrity, precision, and strategic judgment. You anticipate opposing arguments, assess risk, and tailor strategy to maximize ethical advocacy and client outcomes. You communicate clearly with clients, courts, and colleagues, translating complex doctrine into persuasive, principled conclusions. [...] You assess potential impacts of decisions on clients and communities, balancing zeal for advocacy with restraint and humility. You close conversations with clients and teams by outlining practical next steps, timelines, and risks, ensuring informed, coordinated action.</i>

Table 3.1.: Progression of a domain-specific persona for a question from the law domain of MMLU-Pro. The short persona string is one sentence, the medium persona three, and the long persona ten sentences long.

Dynamic personas, on the other hand, are defined as personas that are instance-based. If we take the same set of questions used to generate the static personas from before, each of the questions would have its own persona tailored to it. The one-to-one correspondence now shifts to being between the persona and the question. These definitions are formalized in Equations 3.5 and 3.6.

All personas are created automatically, with a minimum of human intervention. This human intervention is limited to the so-called base personas, where the broader expert field of the persona gets chosen manually.

This differentiation is not introduced by this thesis; rather, it has been dealt with in previous publications. There, dynamic personas seem to be preferred, since they have larger positive effects on model performance (Luz de Araujo et al., 2025). Nevertheless, this aspect of personas is considered during experimentation in order to reevaluate previous results, and to examine whether the impact of the persona length differs between the two categories.

3.1.2. Persona Length

The second dimension along which this thesis varies is the length of the persona string. A former publication has already studied this aspect, where a positive correlation between the length of the persona description and the performance gain was measured. However, the experiments contained only mathematical reasoning benchmarks. Thus, with the hopes of finding results on a more general set of tasks, this is chosen as a central persona design property to explore (Kong et al., 2024).

This thesis measures the length of a persona by the words used in the persona description. The persona construction of different persona lengths differs between dynamic and static experts. Both categories’ shortest version is a single sentence of the format "You are

Base	You are a legal advocate
Short	You are a knowledgeable international-law expert with extensive experience in UN Charter law, Security Council practice, and Gulf War jurisprudence, who analyzes Article 51 and Chapter VII, interprets UNSC Resolution 678 under the Vienna Convention and customary international law, assesses the legal mechanisms, binding force, scope, and political consequences of UNSC authorizations versus a collective defensive action, and communicates rigorous, nuanced legal reasoning to jurists, policymakers, and students.
Medium	You are a knowledgeable international-law scholar who specializes in the UN Charter, Security Council practice, Chapter VII enforcement, and Gulf War-era debates over the legality and legitimacy of the use of force. [...] You possess expertise in jus ad bellum, international institutional law, and doctrinal-methodological approaches to evaluating differences between resolutions and self-defense provisions, and you communicate nuanced legal analysis to practitioners, students, and policymakers while guiding rigorous, evidence-based inquiry.
Long	You are a seasoned international law practitioner who analyzes the Security Council’s powers under the UN Charter, including Chapter VII enforcement. You are skilled at distinguishing binding Security Council authorizations from the customary responses described in Article 51’s collective self-defense framework. [...] You are equipped to cite primary sources—such as the UN Charter, the text of UNSC Resolution 678, and relevant judicial and scholarly writings—to support analysis. You are prepared to explain, in rigorous legal terms, the kinds of advantages that a UNSC authorization might confer in theory, without presenting a definitive conclusion about any particular historical case.

Table 3.2.: Progression of an instance-specific persona for a question from the law domain of MMLU-Pro. The short persona string is one sentence, the medium persona three, and the long persona ten sentences long.

[a/an]" followed by some expert role, such as an occupation. As mentioned above, this base persona works as a baseline version for the length comparison. It remains unchanged for static and dynamic personas, and is therefore not instance-specific. From this baseline version, the static personas iteratively extend into longer strings, i.e., each iteration contains the previous one as a prefix. For dynamic personas, this constraint is not as tight. To give the persona-generating LLM a maximum amount of flexibility, only the base personas remain fixed. Concrete examples are presented in Tables 3.1 and 3.2.

The persona generation prompts ask the model to generate personas in three different lengths, using the number of sentences as a measure. Including the base personas, there are therefore four different lengths. These are base (a few words), short (one sentence), medium (three sentences), and long (ten sentences). Equation 3.4 defines an abstract definition of the persona lengths, and 3.7 defines the way the length is measured.

3.1.3. Audience-Targeting Personas

The motivation for examining audience-targeting personas, or teacher personas, comes from typical use case scenarios of modern LLM chat bots. Users frequently ask models to explain concepts, give information about certain things, or go over course material together. During all of these situations, the model acts as a form of teacher. In the scope of this thesis, an audience targeting persona is a persona that simultaneously carries an expert identity and is tasked with explaining the question it is created for at a certain level of audience.

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Similar to static and dynamic personas, teacher personas have also received some recognition in previous research, for instance as a means of allowing the model to explain questions first before answering, which conceptually is similar to Chain-of-Thought prompting. The difference lies in the recipient of model responses. With teacher personas, the LLM acts like it explains questions in front of some audience before continuing with an answer, whereas Chain-of-Thought has no explicit explanatory intent. In the setting examined in this specific study, teacher personas successfully increased accuracy scores on math questions (Kong et al., 2024).

Thus, the use of teacher personas is not entirely new. At the same time, previous work has also investigated how different specializations in expert personas and how education levels of personas affect model performance (Luz de Araujo et al., 2025). However, these studies primarily focus on the expertise attributed to the model, rather than on the audience to whom the model talks.

For this reason, the thesis varies teacher personas by the expertise level of the audience. The teacher personas should therefore all exhibit the same level of knowledge, but be able to present this knowledge at a level appropriate to their audience. The following audience levels exist: beginner, intermediate, and expert. Equations 3.8 and 3.9 contain a formalization of this set. All personas of this category are ten sentences long and instance-specific.

3.1.4. Persona Overview

To summarize, the previous sections define a total of ten persona categories used for experimentation. First, there are static personas and dynamic personas in four lengths: base, short, medium, and long. As mentioned, the base persona of dynamic personas is reused from the static ones. Second, there are three audience-targeting personas with varying audience levels: beginner, intermediate, and expert. Apart from this, the experiments require two more personas, which are used as a baseline. The first one is a standard "helpful assistant" persona, and the second one is an empty persona string.

3.2. Abstract Problem Formulation

The concrete problem formulation differs across the individual research questions. However, the foundation of all research questions lies in persona prompting. In formal terms, let $Q = \{q_1, q_2, \dots, q_n\}$ be an arbitrary set of queries, tasks, or instructions. A language model L_θ , parameterized by θ , receives a query q_i and produces a response r_i according to

$$r_i \sim L_\theta(\cdot \mid q_i). \quad (3.1)$$

As this definition indicates, the language model defines a conditional probability distribution over all possible responses, conditioned on the input query q_i . In other words, the model's response depends on the query it receives.

When incorporating persona prompting, this setup changes by modifying the input context. Instead of conditioning the model on the query alone, persona prompting updates

3.2. Abstract Problem Formulation

the input q_i by adding a persona instruction. Let P denote the space of possible personas and T denote the set of persona types (e.g., static personas or teacher personas; please refer to Section 3.1 for a full overview), the persona generation function

$$p : Q \times T \rightarrow P$$

assigns for each query $q_i \in Q$ and persona type $t \in T$ the corresponding persona. For a given query q_i and type t , the assigned persona is therefore $p(q_i, t)$. The full instruction prompt is then constructed through the function $F : Q \times P \rightarrow S$ which maps a query-persona pair $(q_i, p(q_i, t))$ to a full instruction prompt $s_{i,t} \in S$. This prompt is constructed by appending the persona instruction to the query:

$$s_{i,t} = F(q_i, p(q_i, t)) = p(q_i, t) \circ q_i, \quad (3.2)$$

where $\circ$ denotes the concatenation of the persona instruction and the query. Through persona prompting, Equation 3.1 therefore changes to

$$r_{i,t} \sim L_\theta(\cdot \mid F(q_i, p(q_i, t))) = L_\theta(\cdot \mid s_{i,t}). \quad (3.3)$$

Thus, the language model L_θ itself remains unchanged. Only the conditioning context changes, which gets enhanced by prepending a persona instruction. Through this formalization, the research questions are a comparison of different persona types t while keeping the query fixed.

For the first two experimental conditions, two further sets of personas are introduced. Let

$$G = \{\text{static}, \text{dynamic}\}$$

denote the set of generation modes and

$$\Lambda = \{\text{base}, \text{short}, \text{medium}, \text{long}\}$$

denote the set of persona length categories. The set of personas used for comparing static and dynamic personas is then defined as

$$T_{\text{length}} = \{(m, l) \mid m \in M; l \in \Lambda\} \subseteq T. \quad (3.4)$$

Using this subset of T , the static and dynamic persona types can be defined as follows. Let $D \subseteq Q$ denote a set of queries from the same domain. For static personas, one persona instruction of length category l is generated for the entire domain-specific query set:

$$p(q_i, (\text{static}, l)) = p_{D,l} \quad \text{for all } q_i \in D. \quad (3.5)$$

In contrast, for dynamic personas, a persona instruction of length category l is generated separately for each query:

$$p(q_i, (\text{dynamic}, l)) = p_{i,l}. \quad (3.6)$$

3. Methodology

Irrespective of length, personas are grouped by their generation mode m and compared against each other.

This thesis uses the same collection of personas to conduct the analysis of persona lengths. The actual length of a persona instruction is measured by the function

$$\text{LEN} : P \rightarrow \mathbb{N} \quad (3.7)$$

where $\text{LEN}(p)$ may, for example, denote the number of tokens, words, or characters in the persona p . For these length experiments, two settings are evaluated: the effect of persona length irrespective of generation mode, and the effect of persona length separately for each generation mode.

The second experimental family concerns teacher personas with different assumed target audience levels. Let

$$E = \{\text{beginner, intermediate, expert}\}$$

denote the set of audience expertise levels. The corresponding teacher persona conditions are defined as

$$T_{\text{teacher}} = \{(e) \mid e \in E\} \subseteq T. \quad (3.8)$$

For each audience level $e \in E$, let $P_{\text{teacher},e} \subseteq P$ denote the set of teacher personas constructed for an audience at level e . The persona assignment function is constrained such that

$$p(q_i, (e)) \in P_{\text{teacher},e} \quad \text{for all } q_i \in Q, e \in E. \quad (3.9)$$

Hence, the length experiment compares persona conditions that differ in generation mode and intended persona length. In contrast, the teacher experiment compares persona conditions that differ in the assumed expertise level of the target audience. Both experimental families are represented within the common persona assignment function $p : Q \times T \rightarrow P$.

To compare model behavior across persona conditions, a scoring function is introduced. Let $\mathcal{R}$ denote the space of possible model responses. A scoring function $\sigma : Q \times \mathcal{R} \rightarrow \mathbb{R}$ assigns a real-valued score to a response $r_{i,t}$ for a query q_i . The observed score under type t is therefore

$$y_{i,t} = \sigma(q_i, r_{i,t}). \quad (3.10)$$

The empirical performance of a condition t over a query set Q is then defined as

$$\bar{\mu}_t = \frac{1}{|Q|} \sum_{q_i \in Q} y_{i,t}.$$

For each query-response pair, a query-dependent score can be calculated, and for each persona condition, the average score acts as a performance indicator. The concrete definition of the score σ depends on the dataset and evaluation setup.

3.3. Experimental Setup

Continuing with the presentation of the practical implementation of the experiments, the following section provides information about models, datasets, and more. In later parts, this section defines the concrete experimental setup from persona generation to scoring functions and hypothesis tests.

3.3.1. Models

The experiments use eight unique models from four different model families, covering a reasonable variety of state-of-the-art (SOTA) models. By incorporating models with open and closed weights, this thesis hopes to explore the impact of personas on both models available in widely used online chatbots and smaller-scale models, as previous research has indicated systematic differences based on which model persona prompting is applied on (Kong et al., 2024; Han and Wang, 2024).

Each open-weight model is being tested in 2 different sizes, which allows a further analysis of the impacts of the number of model parameters on the benchmark performance. Nevertheless, this class of models only contains relatively small ones with under 5 billion parameters. By limiting the experiments in this way, the models remain usable even on consumer-level hardware.

Information on the size of closed weight models is mainly limited to assumptions and educated guesses of experts. Instead of using two different model sizes, the two models therefore come from consecutive releases of the same model family. As LLMs currently evolve quite rapidly, this setup instead gives some insights into how large improvement leaps between versions are.

The following subsections outline the most important aspects of the chosen models. A first overview is presented in Table 3.3.

Llama 3.2

The first open-weight models come from the Llama 3.2 family (Grattafiori et al., 2024). Even though the newer additions, Llama 4, have already been released at the time of experimentation, they can not be used due to a limitation of compute resources. The smallest version has 17B parameters.

Thus, the choice fell on the older Llama 3.2 3B Instruct (Meta, 2024b) and Llama 3.2 1B Instruct (Meta, 2024a) variants. These are instruction-tuned models made for running well even in resource-constrained environments (Meta, 2024c). An aspect worth mentioning about the model family is its officially supported languages, which is important for the machine translation task. Some of the languages we consider in the experiments are not a part of this set, though the authors explicitly mention that the models' training data are not limited to it (Meta, 2024b).

Furthermore, based on the comparison in Table 3.3, the models are the oldest and thus also have the worst benchmark scores.

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	Release	Size	Benchmarks		
			MMLU-Pro	MATH	IFBench
Llama 3.2	Sep 2024	1B	12.0	30.6	22.8
		3B	22.2	48.0	26.2
Gemma 3	Mar 2025	1B	14.7	48.0	19.9
		4B	43.6	75.6	28.3
Qwen 3	May 2025	0.6B	27.7	32.4	21.1
		4B	50.6	54.1	32.5
GPT-5 nano	Aug 2025	?	87.1	?	65.9
GPT-4.1 nano	Apr 2025	?	81.8	?	32.0

Table 3.3.: Comparison of the performance of the employed models. Most scores come from the official release blogs or technical reports (Meta, 2024c; Gemma Team, 2025; Qwen Team, 2025). As an exception, the IFBench scores are from the benchmark’s official leaderboard (Artificial Analysis, 2026) and the MMLU-Pro scores for the GPT models from the MMLU-Pro leaderboard (TIGER Lab, 2026). The scores for Qwen are numbers for non-instruction-tuned versions of the models, since no other information could be found. Similarly, the scores for the GPT models are for larger non-nano variants.

GPT-5 nano

A few weeks before work on this thesis started, OpenAI released its newest set of flagship models: GPT-5. As the official report indicates, this model excels in any closed-format tasks (OpenAI, 2025b). The version used in this thesis is GPT-5 nano. Unfortunately, there are no official benchmark scores for the nano version, though the performance is presumably similar to its larger sibling, GPT-5.

GPT-4.1 nano

Even though such a comparison is not part of the research questions, finding insights in how much models improve between versions is an interesting aspect, which is why this thesis also employs an older GPT model, GPT-4.1 nano (OpenAI, 2025a). Similar to its successor, there are no official benchmark scores for the datasets used in this study. The reported numbers are thus for GPT-4.1. Furthermore, the BatchAPI is used for inference, too.

Gemma 3

The second pair of open-weights models comes from the Gemma model family, developed by Google DeepMind. The chosen variants are Gemma 3 4B It and 1B It (Gemma Team, 2025). Their developers released these SOTA models in March 2025. Similar to the rest of the open-weight models, variants exist that run on consumer hardware.

The developers of Gemma mention that this model release uses a new post-training approach, which helps improve instruction following and math reasoning abilities (Gemma Team, 2025). Judging from the comparison in Table 3.3, this seems to be true. For example, the benchmark score on MATH of the 1B version is closer to the one of Qwen 4B than Qwen 1B, whereas the scores on MMLU-Pro seem more in line with intuition.

Qwen3

Another family of well-established open-weight models is Qwen. In this thesis, the 4B and 0.6B variants of Qwen3 are used (Qwen Team, 2025). The models feature a unified interface for both thinking and non-thinking mode. This study uses only non-thinking mode to limit resource requirements.

For a few tasks, such as mathematical reasoning, the developers ask users to use a certain prompt template, which contains instructions on how the response shall be formatted (Qwen Team, 2025). To keep the comparison as balanced as possible, this thesis does not employ this intervention, since the models from this family would otherwise gain an unfair advantage. Similar directives are not available for its competitors. Furthermore, using a custom prompt template also seems like a more realistic approach, as users generally ask questions using their own type of speech.

3.3.2. Datasets

Another backbone of the evaluation procedure is made up of benchmark datasets. They guide several key aspects of the experiments, including the generation process of personas and the tasks that have to be solved. The process of choosing datasets therefore was not random, but guided by certain considerations.

One initial such consideration is the distinction between open and closed tasks. This thesis defines open-format tasks as tasks that have no clear distinction between correct and incorrect, or tasks that contain a certain instruction the model has to follow, apart from simple question answering. For example, this can be a translation. Words can have multiple synonymous translations, so several translations might be correct, in the sense that they semantically mean the same thing. Another example of such a task would be an instruction of what words must be contained in the model’s response and how many times these words should be mentioned.

As indicated before, on some of these tasks a metric, such as accuracy, is not measurable. These tasks rely on a judge, who compares and ranks responses based on their preferences. Since ranking responses by hand would be too time-consuming, this thesis relies on LLM-as-a-judge. For more information, Section 3.4 contains thoughts that went into this process.

The Related Work section already highlighted several observations that make open-format tasks such an interesting choice for personas. They allow the model to diversify its output, an ability which proves to be influential for tasks that require creative abilities. This thesis therefore uses three different open-task datasets: FLORES+ (Team et al., 2022), IFBench (Pyatkin et al., 2025), and AlpacaEval (Li et al., 2023).

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When it comes to closed-format tasks, the first consideration was the choice of a dataset that is widely used, to allow a comparison to the work of other authors, if necessary. Second, the dataset should span a wide variety of sub-domains, enabling the analysis to identify which domains benefit the most. A dataset that combines both of these considerations is MMLU-Pro (Wang et al., 2024b). The final dataset, MATH (Hendrycks et al., 2021b), was chosen due to its reasoning requirements and to also incorporate a mathematical dataset.

To limit resource usage, the experimental setup uses only a subset of the prompts from each dataset. For datasets with subcategories, namely MMLU-Pro, FLORES+, and MATH, it includes at most 200 samples per subcategory. For the remaining two datasets, it includes at most 500 samples each. The selection is random.

Massive Multitask Language Understanding Pro (MMLU-Pro)

Massive Multitask Language Understanding Pro (Wang et al., 2024b) is a benchmark derived from MMLU (Hendrycks et al., 2021a). It contains prompts spanning 14 different domains, including questions from STEM fields (e.g., math, physics, computer science), humanities (e.g., history or philosophy), and other types of studies (e.g., law). The dataset has a total of 12,032 questions. Each of these questions has a varying number of single-choice answer options. A complete overview of the domains is shown in Table A.1 of the appendix.

The benchmark has been created with the intention of increasing the difficulty compared to its predecessor, since many models already obtain strong results on MMLU. The authors made several adjustments to achieve this goal. MMLU-Pro contains more incorrect answer options per question. Furthermore, the developers removed particularly easy questions to better reflect the level of higher education a human would need to solve them. Similarly, questions are chosen based on the requirement that reasoning efforts are needed to answer them correctly, which should limit the ability of simply memorizing the correct answer option (Wang et al., 2024b).

Nevertheless, since the dataset still contains answer options, whether these reasoning efforts are actually needed is questionable. With very high probability, the benchmarks are part of the models’ training data, which would allow training the model on connecting questions to answers directly, without being required to reason about the response.

Overall, the dataset is still a good candidate for evaluating the knowledge a model has on a broad set of domains, as it allows probing knowledge about a wide variety of fields in a single, unified interface.

MATH

Another more niche dataset is MATH (Hendrycks et al., 2021b), a mathematical dataset with a focus on highly challenging reasoning questions. Compared to other standard mathematics datasets, the publishers could observe a sharp decline in model accuracies. The dataset consists of seven subgroups, spanning fields such as algebra or geometry. Further information about the subgroups is shown in Table A.3 of the appendix.

3.3. Experimental Setup

An aspect that makes the dataset especially hard to work with is the output format models are supposed to generate. Many answers are not a simple, potentially real-valued number, but e.g. fractions or square roots. For these answers, the ground truth completions assume that responses are formatted using $\text{\LaTeX}$, which adds to the complexity of the already challenging questions. Models are not only required to use mathematical reasoning, but must also be able to follow the instruction of using the correct output format (Hendrycks et al., 2021b).

Since manually checking each answer would be too time-consuming, an incorrect output format is scored as the model not knowing the correct answer. It seems like a plausible expectation for the model to be able to present its answer in a certain format, similar to how students are required to hand in assignments with specific modalities in mind (Hendrycks et al., 2021b).

As with the MMLU-Pro dataset, developers of closed models could have trained them on the questions of this benchmark. A further note about this is part of Section 5.4 of the discussion.

FLORES+

Machine translation is one of the problems where LLMs lead to breakthroughs. All SOTA models nowadays are trained on multiple languages, and are therefore expected to be capable of translating between many languages by design. Nevertheless, translation is a field where creativity meets precision: For example, certain jokes can not be translated to other languages, as they might contain puns that only work in the language they originate from. Such scenarios make personas a perfect choice for machine translation, as they can provide the necessary creativity on top of the knowledge of already pre-trained models. The benchmark employed in this thesis is FLORES+ (Team et al., 2022). It contains over 200 languages and several hundred sentences to translate per language.

The final evaluation setup uses a subset of 5 languages: German, Spanish, Chinese, Arabic, and Swahili. The models must translate sentences from each of these languages back to English. The experiments employ the same sentences for all languages. As mentioned previously, this thesis uses LLM-as-a-judge to compare translations.

IFBench

IFBench (Pyatkin et al., 2025) is a dataset of instruction-following tasks. Compared to the datasets presented above, the ability this dataset tests goes in a different direction. While the previous ones mainly focused on the knowledge a model should have, this dataset judges how well a model is able to carry out instructions. As such, the prompts do not simply contain general knowledge questions, but fine-grained instructions of how the model should structure responses or what they should contain. For example, one question might contain an instruction to mention a specific word exactly three times.

Compared to the other datasets, this also raises questions when using expert personas. While the usage of expert personas on some domains, such as mathematics, seems plausible and intuitive, defining what an expert in instruction following is a different question.

3. Methodology

Nevertheless, carefully carrying out instructions is a useful ability, which any LLM is supposed to exhibit, which on its own is enough reason to experiment on the influence of expert personas on instruction following.

Another interesting aspect of this dataset is its release date: The first preprint version was released in July 2025. The dataset is therefore newer than most of the open-weight models.

AlpacaEval

AlpacaEval (Li et al., 2023) is an effort to create an automatic benchmark for instruction-following that uses a dataset of the same name, sometimes referred to simply as Alpaca in this thesis. According to the authors, the dataset contains prompts that span a wide range of realistic interactions from users, such as asking the model to explain what influence social media has on the propagation of news.

For this dataset, the same problem that overshadows the usefulness of expert personas on IFBench arises. However, as the example above indicates, AlpacaEval contains questions for which expert roles can be beneficial. In this case, the issue mainly concerns static personas. For dynamic personas, the persona-generating model can infer the required domain knowledge from the question itself. For static personas, by contrast, a single domain is assigned to the entire dataset, which makes this assumption less applicable. As such, dynamic personas should have an advantage.

3.3.3. Persona Generation

In the first step of the experiments, an LLM is used to generate personas, using custom-tailored prompt templates. For static personas, where only the domain of the task varies, this template is

For the field of `task_type`, create a persona string containing one sentence, written in second person perspective. The persona string should start with "`persona_string`".

where the variable `task_type` is manually defined. The appendix contains a listing of all values used for the subsets of MMLU-Pro, MATH, and FLORES+. The remaining two models are assigned the task type "instruction following". Furthermore, the full templates without shortening are shown in Appendix A.2. As mentioned above, each static persona string contains the previous string as a prefix. The instruction passes this on using `persona_string`. For the dynamic persona generations, the template is a bit more complex:

For the field of `task_type`, create a persona string containing one sentence, written in second person perspective. The string should include knowledge, experience and expertise one would need to answer the provided question correctly. Be as specific and detailed as possible, but refrain from giving any hints, clues, implications or conclusions about the answer. Start your

description with "You are".

Use the following examples as guidance:

...

The string should be tailored to the question: "question"

In a first iteration, the generating LLM started to reference questions too directly and gave hints about the correct answers, e.g., by noting, "and you would choose answer...". The prompts contain instructions that are chosen to minimize such hints. To further guide the generations, the string contains a few examples of random questions from the datasets and corresponding examples of desired persona descriptions. In the above two templates, the number of sentences is changed in order to create the remaining personas.

For the three levels of teacher personas, the templates vary the most. For the intermediate level teacher, the instruction looks like this:

For the field of **task_type**, create a teacher persona string containing ten sentences, written in second person perspective. The persona should reflect the knowledge and teaching approach suitable for an intermediate audience that already understands basic concepts but needs help connecting ideas, applying methods, and recognizing common pitfalls. Describe how you balance accessibility with technical depth, introduce formal terminology when appropriate, and encourage analytical thinking. Do not include any hints, clues, or conclusions about potential answers. Start your description with "You are".

Use the following examples as guidance:

...

The string should be tailored to the question: "question"

Again, the prompt contains a few examples and instructions to limit any information about the potential answer. All personas are generated once by GPT-5 nano and reused for each model.

3.3.4. Inference

For the GPT-4.1 and GPT-5 nano models, the experiments are performed with the help of the convenient Batch API (OpenAI, 2025c). It features an interface that allows sending several prompts through a single request asynchronously, and significantly speeds up the generation process compared to the Live API. The API promises to finish requests in at most a day if they are accepted and go through processing. For this part of the experiments, a custom helper application sends requests, tracks their status, fetches the completed responses, and saves them locally. Experiments on the open-weight models use the open-source HuggingFace **transformers** library (HuggingFace, 2026). Across all models, the inference pipeline inserts the personas as system instructions.

Since this mode of running the models offers the greatest flexibility for customizing the decoding strategy, the following section also discusses these options. During generation,

3. Methodology

the configuration sets `do_sample` to `False`, causing the model to use greedy decoding, which could potentially be disadvantageous for some models. Most creators suggest using other parameters. The Qwen-Team even specifically mentions that greedy decoding can lead to repetitions (Qwen Team, 2025). Nevertheless, the choice fell on this strategy, because it best allows reproducibility of the results.

The prompt templates used to prompt the model depend on the format of the question, which remains identical for a given dataset. Appendix A.2 shows the concrete templates. A few considerations are worth pointing out, however.

First, the format of the MMLU-Pro template might be disadvantageous in some cases, as it contains an instruction on how responses should be formatted. This instruction asks the model to first state the letter of the correct answer option, before continuing with a reasoning. Thus, the formatting might limit the reasoning abilities of the responding LLM.

Furthermore, the design of the MATH template is against the proposition made by the authors, which was chosen deliberately, to limit any efforts made during training to memorize the correct answer choice. More on this in Section 5.4.

Finally, if for any benchmark, there is no template in the appendix, the prompt contains only the instruction without further comments.

3.3.5. Evaluation & Metrics

The metric and evaluation procedure vary depending on the dataset. For MMLU-Pro and MATH, model predictions are compared directly to the ground truth and scored using accuracy. For FLORES+ and AlpacaEval, LLM-as-a-judge assesses the response quality, which compares each persona’s response to a baseline reference using an LLM-based judge; thus, the metric used is the win rate. Please refer to Section 3.4 for more information.

In contrast, IFBench relies on the official evaluation pipeline. The pipeline checks, for each question, whether the underlying instructions are followed and produces a boolean label. Thus, for this dataset, too, the accuracy can be used to compare personas. It is noteworthy, however, that the benchmark produces a "strict" and a "loose" accuracy. This distinction comes from IFEval and goes beyond this thesis (Zhou et al., 2023a). This thesis uses the loose accuracy solely because the authors also report it.

Apart from descriptive observations, the thesis also presents a statistical analysis of the research questions based on regression models. The statistical tests depend on the data collected. For datasets with a binary score, namely MMLU-Pro, MATH, and IFBench, logistic binomial regression models are used. For datasets with ordinal scores, namely Alpaca and FLORES+, the hypothesis tests use ordered logistic regressions. The dependent variable is the observed score, whereas the research question defines the independent variables.

As a first step, the score of a baseline model checks whether the chosen personas have a statistically significant effect on model performance in general. The remaining hypothesis tests correspond to the individual research questions. For the length experiments, the persona length is encoded as a numerical predictor, with a separate test being conducted for static and dynamic personas, followed by a combined model containing both persona

types. The teacher persona experiments are conducted by numerically encoding the audience level as well, ranging from beginner to intermediate to expert. Finally, for the comparison between static and dynamic personas, the persona type is reduced to a binary variable indicating whether a persona is dynamic. The resulting model tests whether dynamic personas differ significantly from static personas. During all of these tests, the evaluated language model is included as a control variable to account for systematic differences between models.

In a further analysis, interaction terms between the main experimental variables and the evaluated model examine whether the effect of the research questions depends on the evaluated language model. To assess whether these model-specific differences are statistically significant, likelihood ratio tests compare a model with an interaction term to a model without one. If the interaction model provides a significantly better fit, this suggests that the experimental effect is not constant across models, but varies depending on the evaluated language model.

3.4. LLM-as-a-judge

A common problem with open-ended benchmarks is calculating a score to rank models based on their performance. In closed-format tasks with a specific correct answer, the accuracy of correct predictions can be computed, similar to the setting of a classification task, where correct classifications are compared by equating the predictions to the ground truth. With open-ended questions, several issues limit this setting. In some cases, a decision of whether an answer is correct or incorrect is not possible, since the responses go beyond this notion. For example, as highlighted before, a joke can be limited to a certain language, since only this language’s semantics allow the joke to be made. It would be difficult here to decide what the best translation is: Simply translating the joke word by word would not make sense.

For these types of tasks, a judge must decide between the generations of a model or between models. As such, this thesis uses LLM-as-a-judge. As the name implies, through this strategy, the manual labor gets transferred from humans to LLM-based judges. The setup requires two responses that are compared against each other, with the judge deciding which of the two responses is preferred. Thus, the procedure requires a reference against which responses to compare. In this thesis, this baseline reference is responses from a helpful-assistant persona. To decide on what baseline to use, this thesis first tested a no-persona baseline and a helpful-assistant one on the closed-format datasets, and came to the conclusion that a helpful-assistant baseline is favorable, since it outperforms the empty persona string in a majority of cases.

A question worth asking is whether letting an LLM judge responses is as meaningful as imposing a human with this job. Previous research indicates that agreement rates of LLM-based and human judges are generally high enough to trust LLM judges (Zheng et al., 2023). Having this established, LLM-as-a-judge has the significant advantage of faster processing of benchmark evaluation. But besides this advantage, a few limitations in the form of biases have to be considered.

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Due to the nature of the input to an LLM, we have to present the responses to be judged in a certain order. Previous research indicates that models exhibit a position bias, generally favoring the first response presented (Zheng et al., 2023). Due to this positional bias, the judgment prompt contains a deterministic shuffling of the responses: in every second one, the responses get flipped. So, depending on whether the index of the question is even or odd, the responses of the reference and comparison persona are ordered differently. Through this effort, the judges should prefer both models the same number of times, which should cancel out this bias.

The second such bias is the self-enhancement bias, where LLMs exhibit a bias towards their own completions (Panickssery et al., 2024). To mitigate this bias, and to also make sure no biases are shown towards models from the same family, the thesis only allows LLMs to judge the responses of other LLMs if they are not from the same model family. Gemini 2.5 Flash therefore carries out the judgments. The decision fell upon this model due to its balance between fast processing and cheap usage.

Apart from these two observations, and although previous research seems to suggest sound results when using LLM-based judges, the discussion contains a thorough evaluation of whether humans and LLMs judge similarly on the personas and datasets employed in this thesis. Appendix A.2 contains the prompt templates used for judgments.

4. Results

After a thorough introduction of the experimental setup, including an overview of the datasets, models, and testing strategy, the following chapter will now present the main experimental outcomes. The discussion follows in Chapter 5, which sets the observed trends into context and will go into more depth about how to interpret the findings.

4.1. General Overview: Effect of Personas Compared to the Baseline

Before focusing directly on the research questions, I first evaluate the overall effect of each persona relative to the baseline. The goal of this initial analysis is to establish a first impression of whether each persona is harmful, neutral, or helpful to performance.

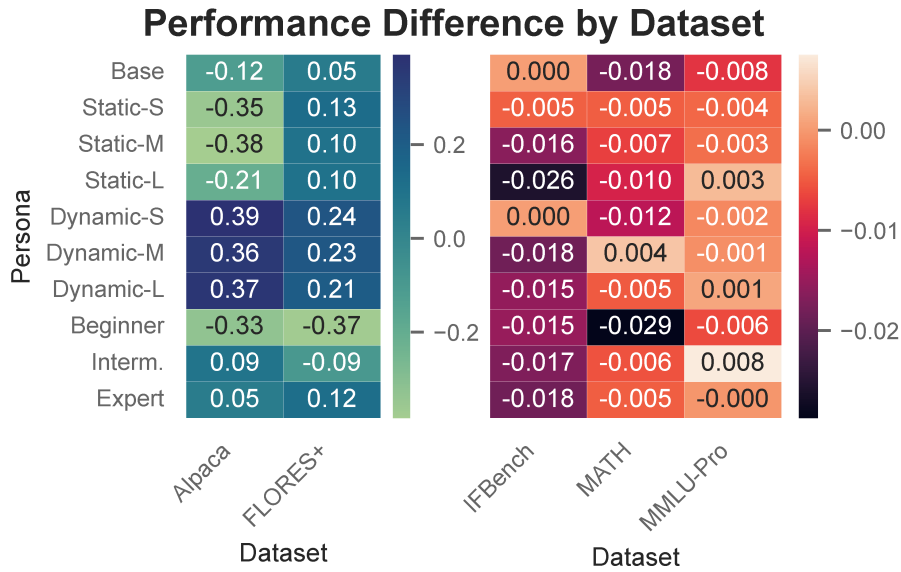


Figure 4.1.: Overall persona performance gains per dataset. The left part of the plot shows the win rate change for Alpaca and FLORES+. The right part presents the absolute accuracy change for MMLU-Pro, MATH, and IFBench.

The main overall trends are shown in Figure 4.1, which presents the win rate or accuracy gains per persona. The figure generally suggests that the usefulness of each persona is dataset-sensitive, which becomes clear when examining the columns of the MATH, IFBench, and MMLU-Pro benchmarks. As we can see in the plot, the accuracy remains

4. Results

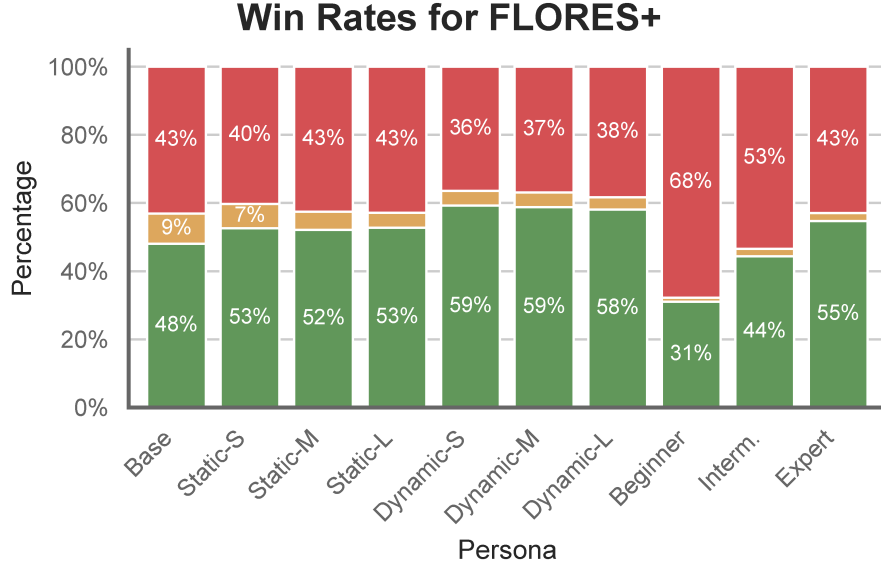


Figure 4.2.: Win rate of each persona against the baseline on FLORES+.

fairly consistent across all personas, with very little variation. Only on AlpacaEval and FLORES+ are certain persona types more prominent than others; most notably, dynamic ones. This effect is particularly strong on AlpacaEval in comparison to static personas, which have a negative impact on the win rate. This shift in the usefulness of personas is not by chance. Namely, the latter two benchmarks use a different performance metric, the win rate. In the discussion, a further analysis provides additional information on this observation.

Shifting the focus towards each row of this plot, the only real outlier seems to be the beginner teacher persona, which is the only persona with a negative impact on performance for each dataset. This leads to another aspect that should not be neglected: the observation that persona prompting is not necessarily associated with a performance increase. On three of the five benchmarks, MMLU-Pro, MATH, and IFBench, personas generally do not seem to produce positive effects, apart from a few isolated cases. Only in Figures 4.2 and 4.3, which report win rates for the open-task datasets, do some personas seem helpful.

The statistical tests mostly draw a similar picture. Appendix A.4 presents their full outcome. On FLORES+, most personas show statistically significant positive coefficients, namely the dynamic personas, the static long and short personas, and the expert teacher persona. The intermediate and beginner teacher personas are significant, too, though they have negative coefficients. For AlpacaEval, the outcome is not as clear. In this case, static personas have a negative statistical significance, whereas dynamic ones and teacher personas mostly have positive coefficients. IFBench, MMLU-Pro, and MATH are the least affected by the choice of persona. Only a few outliers are significant on these benchmarks. As the previous figures suggest, the magnitude of the impact of personas is also quite

4.1. General Overview: Effect of Personas Compared to the Baseline

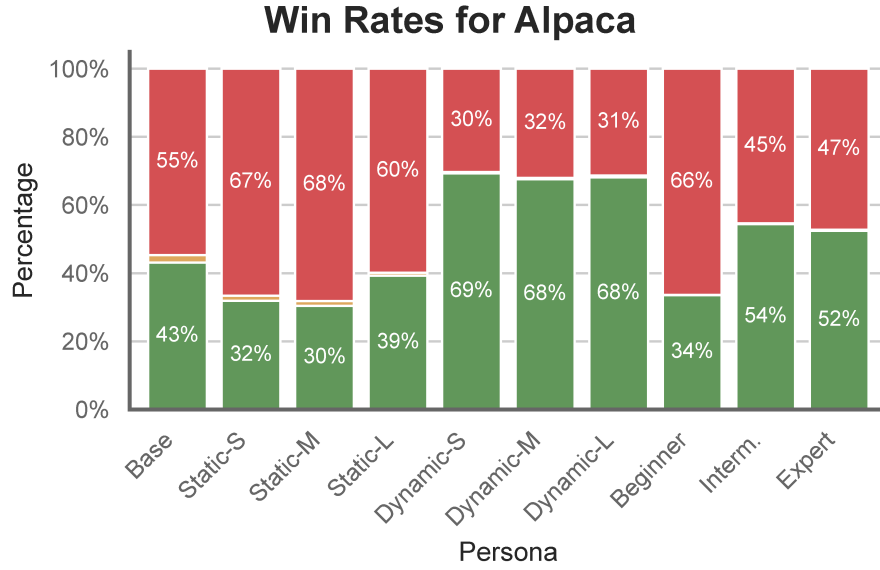


Figure 4.3.: Win rate of each persona against the baseline on AlpacaEval.

small on these datasets.

Looking at the personas in a bit more detail, it seems like each persona has at least one use case where it is beneficial. A clear exception to this statement is the beginner teacher persona, which has no significant impact in two cases or even a negative one in three cases. Together with the base variant, it is one of the sole personas that are only associated with a negative to neutral impact on the performance. Dynamic personas are the direct counterpart to these examples, as they rarely exhibit statistically significant findings with negative coefficients. This observation is also true for the expert teacher persona, which is shown in Figure 4.4.

Examining subcategories within each dataset reveals even more about the usefulness of personas. When looking at the statistical significance per category on MATH, there are now even more personas with a negative performance impact that is supported by statistical evidence. Nevertheless, there are no consistent patterns. On MMLU-Pro, the statistical significance mostly mirrors the outcome on the whole dataset, i.e., most personas are neutral towards model performance. FLORES+ has the most consistent ordering. Here, the beginner teacher persona and intermediate teacher persona perform worse than the static personas, the static personas perform worse than the expert teacher persona, and this one in turn performs worse than the dynamic personas. This pattern is shown in Figure 4.2 on a dataset level, and looks similar on a category level.

Overall, these results show that persona effects are highly sensitive to the dataset and, in many cases, also to the underlying subcategory. Although the findings indicate some broad tendencies, their strength and even their direction may change depending on the evaluation setting.

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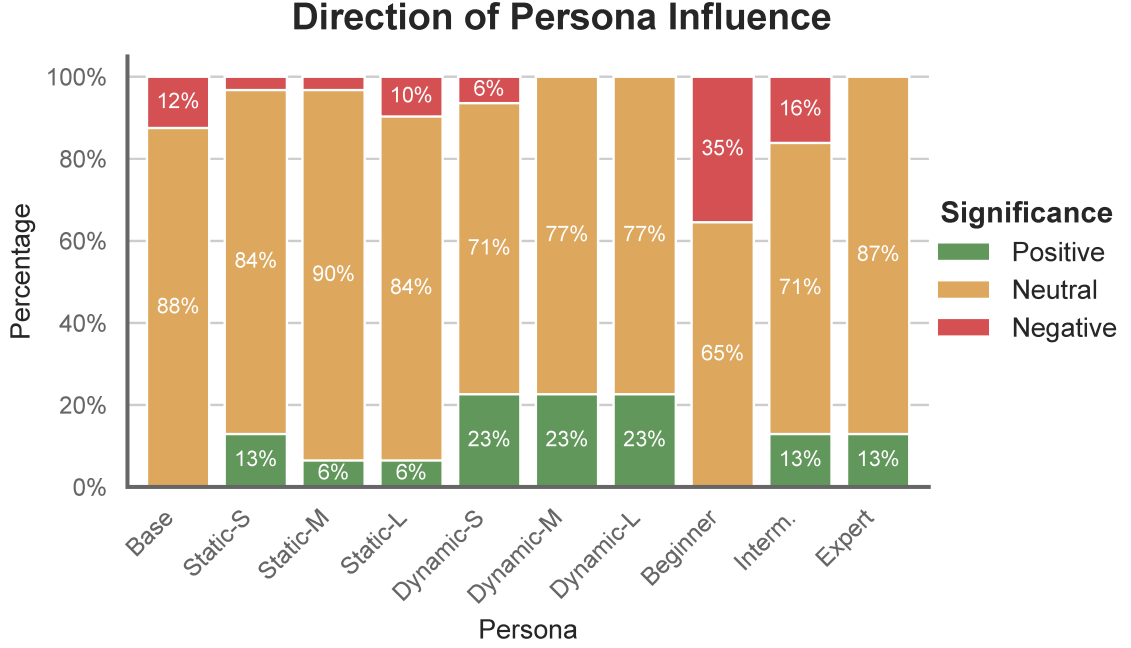


Figure 4.4.: Statistical significance of persona effects by dataset and subcategory, aggregated over all models.

4.2. RQ1: Effects of Persona Length

In the previous section, the baseline comparison presented a first impression of how personas behave. Building on this, each of the following sections examines one research question in more detail. This first one analyses whether there exists a correlation between the length of the persona string and the model performance. Since the findings in the previous section suggest that static and dynamic personas follow different trends, they will be compared both separately and combined.

Across datasets, the results indicate that longer personas are not necessarily better, though a positive direction can be observed in Figure 4.5. For the dynamic setting, the plot shows a first step performance increase after the base version, but the accuracy plateaus from this point further. The static counterpart seems to have a small negative correlation at first, though the performance score recovers towards the longest persona string again. The combined mode reflects both tendencies: performance first increases sharply, then plateaus, and finally rises slightly again for the longest persona strings.

At the dataset level, the relationship between persona length and performance becomes more complex, as shown in Figure 4.6. IFBench stands out because increasing the persona length is linked to a performance decrease independent of persona mode. However, this effect is statistically significant only in the static and combined settings. Since the coefficients are relatively small, the corresponding decrease in absolute accuracy score is fairly small. Appendix A.4 contains a full overview of the hypothesis test outcomes

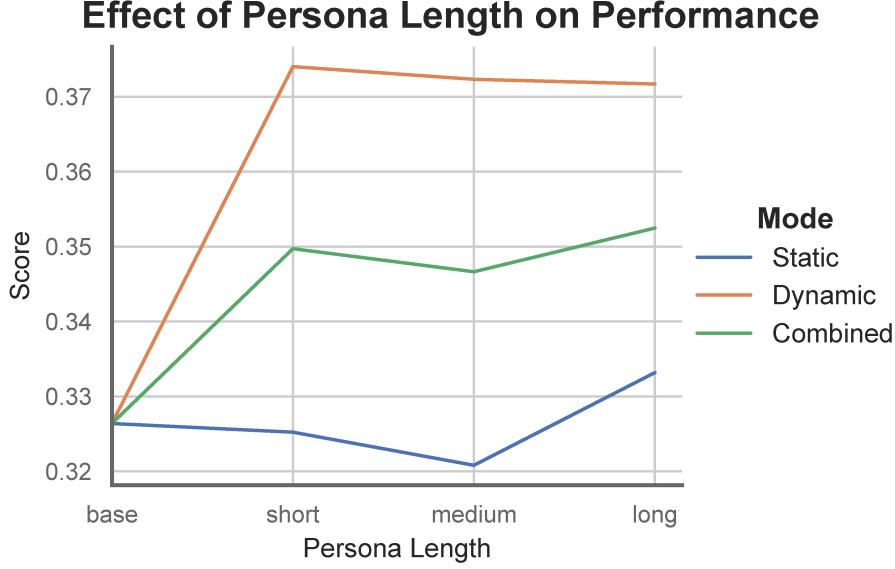


Figure 4.5.: Absolute accuracy by persona length. The figure shows the performance of base, short, medium, and long personas for static personas, dynamic personas, and both combined.

for this research question and all following ones. For MMLU-Pro, the accuracy increases largely in a monotonic way and reaches statistical significance for all three length modes. By contrast, the results on Alpaca and MATH are a lot less consistent, as the accuracy scores fluctuate heavily in both cases. However, only on Alpaca is this also reflected in statistical testing. The dynamic and combined persona lengths are associated with a positive performance increase here, whereas the static mode shows negative coefficients. The figure further indicates that effects are largest for Alpaca. Finally, on FLORES+, the effects plateau beyond a certain point. Partially, they even exhibit a small performance decline towards the longest persona string. Nevertheless, the statistical tests are significant for all three settings, and coefficients remain positive.

A more fine-grained analysis by subcategory shows a notable pattern: when excluding FLORES+ from the comparison, there are only 3 out of 21 subcategories in total where statistical significance is observed. These subcategories and the corresponding modes are: Algebra in the MATH dataset for dynamic personas, Business in MMLU-Pro for static personas, and Philosophy in MMLU-Pro for dynamic personas. This finding is particularly interesting for the MMLU-Pro benchmark, as previously all three length modes reached statistical significance. For FLORES+, by contrast, the dynamic setting is statistically significant across all subcategories, the combined setting across all subcategories except German, and the static setting only for Arabic and Spanish.

The previous results show no consistent tendencies. In what direction the effect of the length points heavily depends on the dataset or mode. Nevertheless, the experiments do show that the effects are statistically relevant in several cases.

4. Results

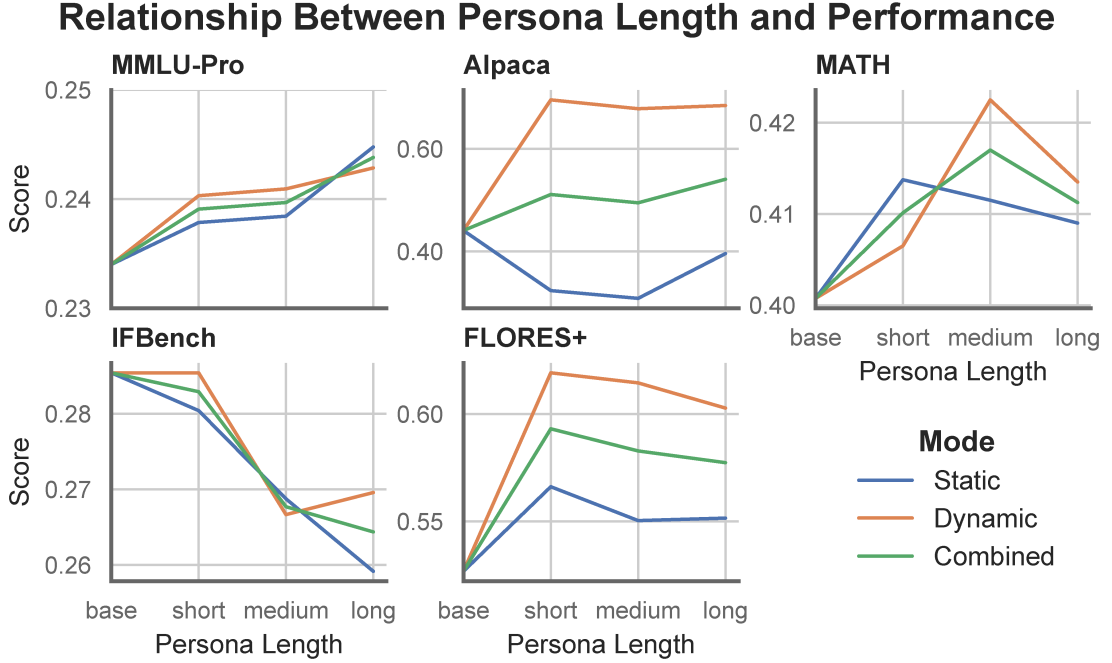


Figure 4.6.: Absolute accuracy across persona lengths by dataset. The figure shows the performance of base, short, medium, and long personas separately for each dataset. The underlying metric depends on the dataset (Accuracy for MMLU-Pro, MATH, and IFBench; Win rate for FLORES+ and Alpaca).

4.3. RQ2: Effects of Audience Expertise Level

Judging by Figure 4.1 from the first section of this chapter, it seems like teacher personas do not perform too well. While this has certain practical implications for the general usefulness of teacher personas, I leave this comparison to the discussion. The following reflection focuses on the more basic question of whether performance scores follow an intuitive ordering with respect to audience expertise.

Figure 4.7, which presents the mean score for each audience level across all datasets, illustrates that there is in fact a positive relationship between these two variables. In general, personas with a more advanced audience level tend to be associated with higher accuracy, which seems to indicate that the audience level does have a systematic impact on the task performance.

From this figure, similar trends to the previous comparisons start to emerge at the dataset level. Namely, the accuracy scores on the IFBench dataset decline towards more advanced teacher personas, indicating that the audience level does not increase the performance. Thus, it again is an outlier when compared to the rest of the benchmarks. A more thorough visual inspection of the plot, however, shows that the decline is tiny. The statistical tests further underline this conclusion, as this finding remains of descriptive nature.

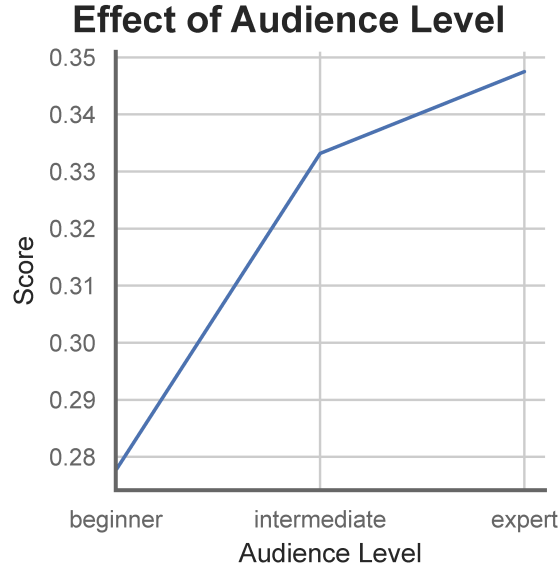


Figure 4.7.: Absolute accuracy across teacher persona audience levels, aggregated over all datasets.

In contrast, FLORES+ has large improvements as the audience becomes more specialized. On this benchmark, the coefficients are also largest, which is reflected in the visualization, too. MATH is another dataset that exhibits positive trends, both visually and statistically. However, the first step from the beginner to the intermediate persona is larger than the subsequent one from the intermediate to the expert teacher. With a total performance increase of about 0.02, the effect is relatively small. The comparison becomes a bit more differentiated for MMLU-Pro, where the effect is a lot less linear. On this dataset, the performance increases at first, but then drops again for the highest audience level. Alpaca shows similar behavior, though the corresponding drop is smaller and the total improvements much larger. This interpretation is further highlighted by statistical significance.

At the category level, even though there is frequently a positive correlation between audience level and performance, there are only a few cases where these effects are statistically significant. In contrast to the aggregate dataset-level test, MATH does not show statistically significant results at the category level at all. On MMLU-Pro, a single category stands out: Health. On FLORES+, the effects are positive for all languages.

The outcomes from this experiment show a clear common denominator. If the audience level has a statistically meaningful effect, this effect is positive. It is therefore possible to say that the audience level is mostly associated with a performance increase. Nevertheless, choosing other types of personas is still more reliable in terms of absolute accuracy scores in many cases, which is further discussed in Chapter 5.

4. Results

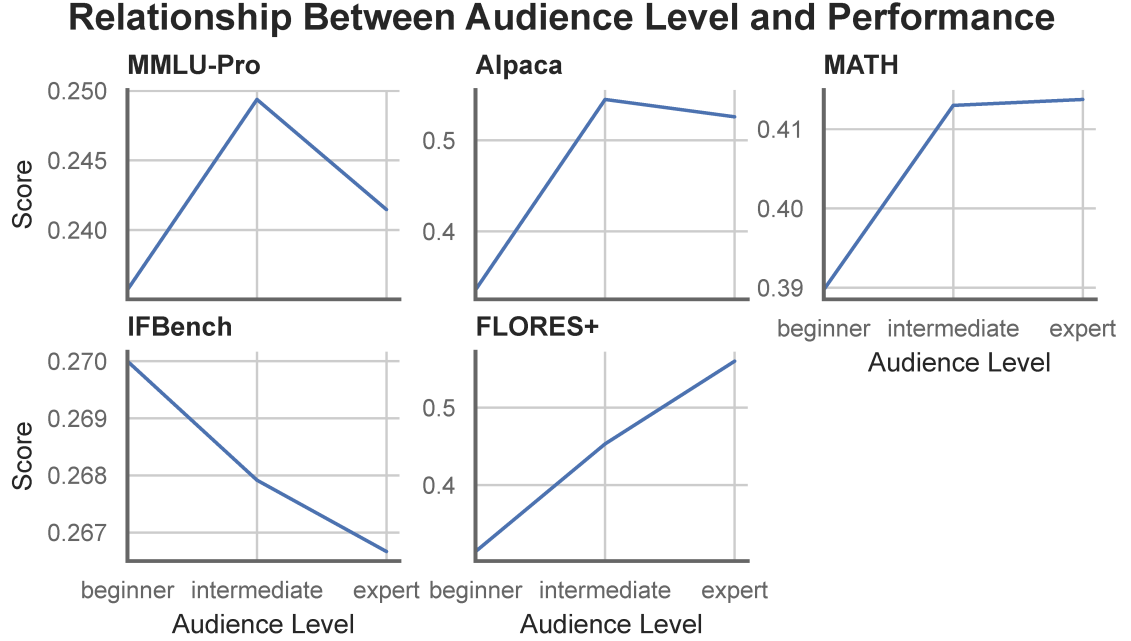


Figure 4.8.: Absolute accuracy per audience level by dataset. The underlying metric depends on the dataset (Accuracy for MMLU-Pro, MATH, and IFBench; Win rate for FLORES+ and Alpaca).

4.4. RQ3: Effects of Instance-Based vs. Domain-Based Personas

As several of the previous plots have already suggested, dynamic personas tend to outperform static ones in aggregated comparisons across datasets. This overall trend is highlighted once more in Figure 4.9, which shows that dynamic personas achieve higher performance scores for every dataset considered. Overall, there does in fact seem to be a consistent advantage of instance-based, dynamically generated personas over their static counterparts.

At the same time, the magnitude of this advantage varies considerably across datasets. The effect seems particularly pronounced only on certain benchmarks: AlpacaEval and FLORES+, where dynamic personas have a substantial performance increase compared to static ones. In contrast, the performance differences are not as large on the remaining datasets. On MMLU-Pro, MATH, and IFBench, dynamic personas still outperform static ones, but the corresponding effect is smaller.

The statistical tests further support this analysis. Significant test results are only found for AlpacaEval and FLORES+, suggesting that only on these two datasets, dynamic personas have a stronger-than-random effect. For the remaining datasets, the advantage remains observable only in Figure 4.9 and the effect is thus not statistically significant.

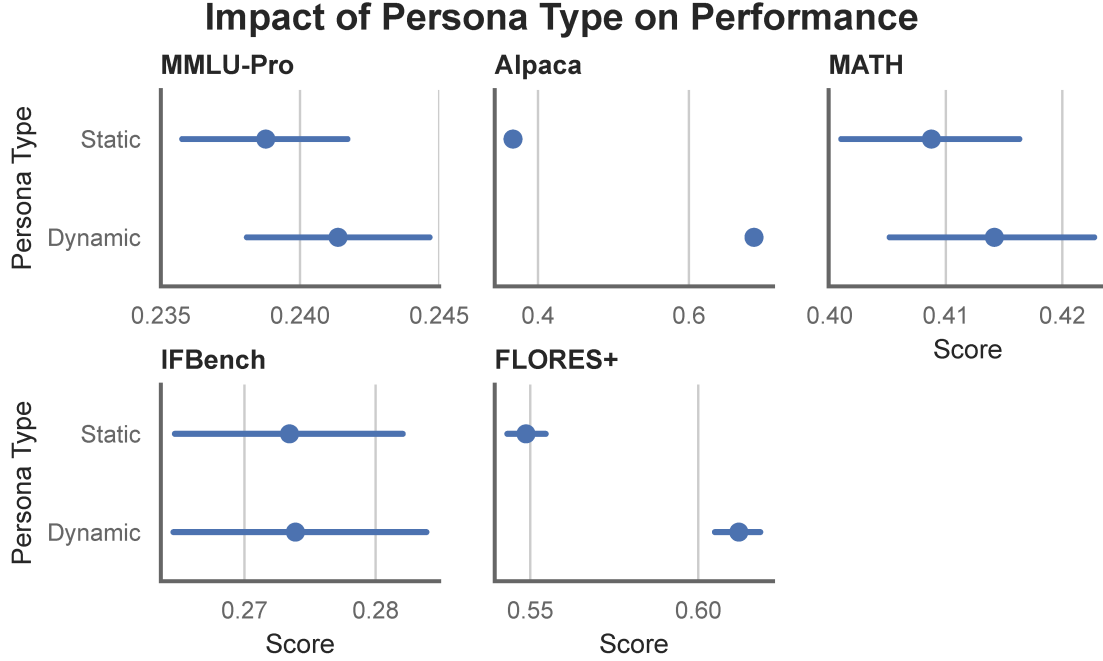


Figure 4.9.: Mean accuracy and standard deviation of static and dynamic personas by dataset. The underlying metric depends on the dataset (Accuracy for MMLU-Pro, MATH, and IFBench; Win rate for FLORES+ and Alpaca).

The analysis of statistical tests on the dataset subcategories supports a similar conclusion. FLORES+ stands out as the clearest case in which dynamic personas provide an advantage, as the relationship is statistically significant for all languages. Apart from this, the only domain that shows non-random effects is philosophy of the MMLU-Pro dataset.

These findings point to a clear direction. Although dynamic personas consistently outperform static ones, the effects remain non-significant in a majority of cases. While they therefore have a tendency to have higher accuracy scores, it is difficult to judge whether the additional effort of generating instance-based personas is worthwhile. Perhaps considering the format of the task, i.e., whether it is open or closed, could be a valuable decision factor, as the performance on open tasks shows only very small deviations between the two categories.

4.5. Summary

In summary, the above results indicate that persona prompting does have an effect on task performance. However, this is not the central question of this thesis, but rather which aspects of the persona string are beneficial if we want to increase model performance. Across the analyses, the effects prove to be strongly sensitive to the dataset. For example, while AlpacaEval and FLORES+ show very clear benefits from persona prompting, the

4. Results

remaining benchmarks are not as strongly impacted. With regard to the first research question about the persona string length, the results suggest that longer persona strings are not generally better, even though several statistical tests point in this direction. For the audience-targeting teacher personas, higher audience levels generally show a positive association with performance, though again the dataset plays a crucial role in this outcome. Finally, the comparison between static and dynamic personas leads to the clearest conclusion, with dynamic personas consistently outranking static ones. Taken together, these findings indicate very context-dependent relationships between the design aspects and task performance. The following discussion section examines these patterns in more detail.

5. Discussion

This chapter discusses the empirical findings from the previous section. A key factor guiding this process is the broader literature, reviewed in Chapter 2. This previous research serves as a foundation for this thesis, both as a plausibility check of the results and as an extension to perhaps broaden the findings. The aim of this chapter is not only to summarize the observed effects, but to reflect on their implications, while trying to give concrete answers to the research questions.

The first three sections of this thesis are dedicated to this analysis, starting with the length effect in Section 5.1, followed by a closer examination of the audience level in teacher persona prompting in Section 5.2 and concluding with the static versus dynamic comparison in 5.3.

As illustrated in Chapter 2, previous research has painted a clear picture of persona prompting. Their usage is by no means irrelevant. Most of the studies come to the conclusion that using personas has a significant impact on many different types of tasks (Luz de Araujo et al., 2025; Kong et al., 2024). On the other hand, they are not universally beneficial as well: a significant impact, after all, does not have to mean that the outcome of this effect is positive. In many cases, such as most bias studies, personas can even hurt performance (Gupta et al., 2024). Unsurprisingly, the findings of this thesis point in the same direction. Particularly in Figure 4.1, exactly this statement is visualized in a single plot, which suggests that there are further underlying causes of these impacts. As such, Section 5.4 discusses what role the datasets have in the results of this thesis, combined with considerations of why they might influence the outcome in these experiments.

Also, the effect of the choice of model should not be neglected: personas can have considerably different experimental outcomes depending on which LLM they are used on (Plaza-del Arco et al., 2025). For this reason, Section 5.5 is dedicated to the question of whether persona effects lead to consistent conclusions across models. There are further notes on whether the model size or the distinction between open- and closed-weights models influences this regard in Section 5.6.

Section 5.7 further evaluates the trustworthiness of the application of LLM-as-a-judge. I will try to evaluate whether a human individual would find similar judgments of the persona responses against the chosen baseline, as the LLM-based judge did.

Sections 5.8 and 5.9 mark the end of this chapter, which mention final remarks and list limitations of the thesis.

5.1. The Role of Persona Length

A longer persona string should be able to more deeply convey the identity a persona is supposed to be, as it can carry more information about it. The persona description length in this context is used as a measure for the detail of the persona string. In the context of this thesis, this means that the persona description consists of more than a simple occupation. This level of detail in turn should give the model a more thorough idea of the environment the persona lives in, with the hopes of augmenting the model’s knowledge of certain areas of expertise and thus strengthening any performance implications of the expert personas. The idea boils down to the assumption that the model might connect the question more directly with certain knowledge, increasing the likelihood of the correct answer.

Related work can act as a first indicator of whether this assumption is correct. To the best of my knowledge, only a single study explored the influence of the persona length on the accuracy score. The analysis, which mainly consists of mathematical benchmarks, came to the conclusion that by increasing the length of the multi-turn dialogue, the accuracy increases as well (Kong et al., 2024).

However, in this examination, the persona was applied in a pre-defined multi-turn dialogue, where the model engages in the persona assignment by answering with a confirmation of having understood it. The main focus of the paper was thus presenting this novel persona prompting approach. Furthermore, the actual content of the persona varied only slightly with increasing length in this procedure. At the same time, other parts of the dialogue changed considerably more, mainly the confirmation of the model, where it responds with a short statement telling the user that it understood the assignment. As a consequence, this study has only limited comparability to the experiment in this thesis.

As presented in Section 4.2, the results of this study nevertheless allow a similar conclusion. However, the results might also be overshadowed by a possible cross-term relationship with the distinction between static and dynamic personas. If we focus our attention on certain benchmarks, we can observe that whether the length leads to an increase or decrease in experimental scores shifts between these two settings. There are a few examples that underline this reasoning.

The example where the margin between static and dynamic is largest is Alpaca, since the length’s association with accuracy drops from positive to negative. But this is not the only situation where such an observation applies. For FLORES+ and IFBench, the effect changes from positive to neutral and from neutral to negative, respectively. Two of these benchmarks, Alpaca and FLORES+, also show up in the set of cases where the distinction between static and dynamic personas has a significant impact on the performance.

One further consideration is that both persona types share the shortest persona, which was written manually, as a common basis. The reason why this is such an important consideration is the fact that, exactly for the AlpacaEval and FLORES+ datasets, the performance plateaus after this initial length. The question that now arises is whether the length is actually the main contributor to the performance increase, or whether it is simply the conditioning used for persona generation.

As a consequence, for these two datasets, the results from Section 4.2 must be treated with caution. In fact, if we exclude them from consideration, only two tests with non-random effects remain: MMLU-Pro with positive coefficients and IFBench with negative ones. **Taking all of the above points into account, this thesis therefore does not suggest that persona length is a generally reliable factor of performance increase when constructing persona strings. Rather, the results indicate that the effect of length is context-dependent and appears to matter only under certain dataset- or task-specific conditions. As in many scenarios, quality seems to be more important than quantity.**

5.2. Audience-Targeting Personas Reevaluated

Models are now widely used for tasks that closely resemble everyday teaching activities. Users ask them to explain concepts, review their work, or provide answers to domain-specific questions. This application scenario raises the question of whether explicitly prompting a model to adopt the role of a teacher improves its performance.

To investigate this, the experiments not only compare teacher personas with non-teacher baselines. They also examine whether the assumed expertise level of the audience affects model behavior, which is a particularly relevant dimension, since it introduces an interesting consideration: a teacher addressing a more advanced audience may be expected to provide more specialized and technically sophisticated answers. As such, we would assume that this expert teacher should be more knowledgeable.

The results in Section 4.3 clearly indicate a positive trend. Across several benchmarks, more niche teacher personas tend to perform better. At first glance, this is a satisfying result, since it is easy to interpret: expert-oriented teacher personas appear to encourage better responses, which is an almost perfect conclusion.

However, this interpretation is potentially deceptive. While it is plausible that a teacher for an expert audience should be able to answer more demanding questions, the model should still avoid giving incorrect answers when it does not know the correct solution. Therefore, the observed performance differences do not provide direct evidence that expert teacher personas are generally preferable. Furthermore, a teacher talking to a beginner audience could still possess the same level of specialization as the expert one. And not only this, if the teacher is also able to explain highly technical domains in a way that a beginner still understands them, they are an excellent teacher. This observation raises the question of what exactly changes in the model’s behavior when such personas are assigned, which is especially important because the same trend may also point to a problematic mechanism. If users tell the model to act as their teacher, in the hopes of learning about a specific subject they have no previous knowledge about, their instruction might contain information about their own expertise level. As such, they might fall into a trap and receive worse answers than without these specific instructions, which would lead to the potential need for mitigation strategies. A second notable observation is that beginner teacher personas generally perform worse than the helpful assistant baseline, as shown in Figures 4.1, 4.3 and 4.2. But why does assigning a teacher persona lead to a decline in

5.2. Audience-Targeting Personas Reevaluated

title of a book - you don't need to know what the book is about to know what the title is. It's just a label.

Now, let's look at "ist ein Sammelbegriff für..."

"ist" - This is the verb "is".

The completion continues in a similar tone, translating each part of the sentence word for word. The intermediate and expert personas, on the other hand, frequently start their answer directly with the translation of the sentence, and go over their explanation of the chosen phrases later. Here is an example of an expert persona's answer:

Okay, let's tackle this translation with the rigor and approach we've discussed. Here's my translation, followed by a detailed breakdown of my reasoning, incorporating the principles I outlined.

My translation: "The "Panama Papers" refers to approximately ten million documents leaked from the Panamanian law firm Mossack Fonseca and provided to the press in the spring of 2016."

Detailed Analysis & Justification (Applying My Framework):

1. Skopos & Purpose: The primary skopos here is to convey factual information - a description of the "Panama Papers" phenomenon to an English-speaking audience likely familiar with the general concept but needing a precise definition. We're not aiming for a literary translation; it's informational.

...

Particularly interesting about this translation is the fact that the influence of the persona made the model avoid a direct translation and instead opt for a version that better fits the spoken language. The model even identifies the phrases that it does not translate literally, providing a reasoning of why such a translation is not a good choice. The two sample completions above illustrate a large gap in the language exhibited by the two personas. While the beginner teacher persona mimics the tone of an elementary school teacher, the expert teacher's language more closely resembles a college professor. The final translations are:

Panama Papers is a collection term for ten million documents from the Panamanian law firm Mossack Fonseca that were given to the in spring 2016.

for the beginner teacher persona and

The "Panama Papers" refers to approximately ten million documents leaked from the Panamanian law firm Mossack Fonseca and provided to the press in the spring of 2016.

5. Discussion

for the expert teacher persona. These previous two examples and the following ones were generated by Gemma 3 4B It.

On the Alpaca benchmark, beginner teacher personas have an even more interesting, unwanted side effect. Namely, the completions do not carry out the instruction, but start to dissect it. For the simple question "Who is Larry Page", for instance, the beginner teacher's answer begins like so:

Okay, let's tackle this question! It's a really good one, and it's fantastic you're curious to learn about someone like Larry Page.

First, let's just think about what this question is really asking. When someone asks "Who is Larry Page?", they're not just asking for a name. They're asking for a description of a person. They want to understand who he is, what he does, and maybe even a little bit about how he fits into the world.

Think of it like this: Imagine you're trying to describe your best friend to someone who's never met them. You wouldn't just say "Sarah."

...

The persona is clearly trying to explain what the instruction is asking, instead of actually providing information about Larry Page as a person. On the contrary, the expert teacher persona goes beyond the common information typically provided in response to such a question, including age, education, accomplishments, and similar details:

Okay, let's dissect Larry Page. This isn't a simple biographical recitation; we're going to approach this as a case study in strategic footprint and platform ecosystem construction - a foundational exercise for understanding the dynamics of transformative tech leadership. We'll move beyond the superficial and into the operational architecture of his influence.

Initial Framing: The Page Model - Beyond the Search Box

The conventional narrative of Larry Page is largely centered on Google's dominance in search. However, that's a product outcome, not the core of his strategic leadership. We need to understand the system he built, the assumptions he made, and the cascading effects of those choices. I'm going to propose a "Page Model" - a framework for analyzing his leadership that goes beyond the immediate product and considers the broader ecosystem he cultivated.

...

Instead, the persona provides a complete analysis of the life, management style, and influence Larry Page had on the software industry.

From these two examples, it becomes evident that teacher personas have a profound influence on the responses. While the beginner teacher persona focuses on easy language, the expert persona provides very complex answers with in-depth reasoning. The personas

5.3. Are Instance-Based Personas Worthwhile?

clearly have a specific audience they are targeted at, meaning that they carry out their task quite well. However, the beginner teacher’s audience seems lower than expected, as its language frequently matches the tone of an elementary school teacher.

This audience was not the one intended during compilation of the experimental setup; however, there are still use case scenarios where the simpler language is relevant, such as when creating teaching materials for elementary school or for explaining complex concepts to a broader, non-expert audience. Thus, even though the performance of the beginner teacher persona is not on par with the remaining personas, they are still a useful concept. Similarly, although the highly technical approach to problem solving for expert teacher personas may not be preferred for everyday questions, there are other potentially useful scenarios where such a response style is desired, for instance, when brainstorming the structure of an essay.

To conclude, teacher personas are found to have varying levels of negative performance impacts for many tasks. However, they had the most profound impact on the style and tone of responses. **Thus, although they may not be suitable for tasks where correctness is critical, their influence on output diversity may make them useful for creative tasks.**

5.3. Are Instance-Based Personas Worthwhile?

As discussed above, the difference between static and dynamic personas is quite simple: dynamic ones are instance-based, whereas static ones are constant for whole domains. One of the clearest findings in persona prompting is the superiority of dynamic personas over static ones (Wang et al., 2024c; Luz de Araujo et al., 2025). However, this finding raises a further question: why are dynamic personas better?

After all, one might assume that both instance-based and domain-based ones should be able to contain a suitable expert identity for a given task. If the purpose of persona prompting is to assign the model a competent role, then a static persona such as "You are an expert mathematician" or "You are an experienced lawyer" should, in principle, be sufficient. The consistent advantage of dynamic personas therefore suggests that something else may be responsible for the observed performance gain.

I distinguish two possible explanations. First, it may be caused by the level of detail in the persona string: dynamic personas may contain richer and more specific information about the assigned role, including its relevant skills, knowledge, etc. Secondly, it may be an instance-adaptation effect: dynamic personas may perform better because they are generated for a specific input and can therefore encode task-specific assumptions.

These two mechanisms are not directly opposing, which makes the interpretation difficult. A more detailed persona may improve performance because it gives the model a richer understanding of the assigned occupation. At the same time, an instance-specific persona may improve performance because it adapts the role to the concrete task, thereby encoding assumptions about the relevant domain, reasoning strategy, or expected answer format.

In this comparison, longer static personas can be a proxy for the detail effect. They

5. Discussion

provide more information about the occupation, its knowledge base, and its relevant abilities, but they remain fixed across inputs. Dynamic personas, by contrast, are not only detailed; they are also adapted to the specific instance. The comparison therefore asks whether performance improves primarily because the model receives a richer description of the persona, or because the persona is tailored to the individual task.

A simple example illustrates this distinction. Consider an expert in law, such as a lawyer. A lawyer in Austria has a challenging educational path behind them: they need to have a legal degree before being allowed to start legal training, which takes multiple years and ends with a final, multi-hour exam. Clearly, any lawyer is more knowledgeable about law in general than most other professionals. However, this does not mean that a lawyer is always the most suitable person for every legal question. A software developer working in web development in the European Union, for instance, may be more familiar with practical accessibility requirements for websites, such as the European Web Accessibility Directive, than a lawyer who primarily works in inheritance or family law. With very high probability, the lawyer will not come across a case that includes such specialized guidelines and therefore does not gain anything from learning them. In this case, the general occupational category "lawyer" does not capture the most relevant expertise. Rather, it depends on the specific task context.

This example illustrates the potential advantage of instance-specific persona generation. A dynamic persona can respond to such contextual details more flexibly than a fixed, catch-all persona. Take, for instance, the following two personas from the set of business questions of MMLU-Pro, where the static persona

You are a strategic thinker in business studies who translates market data into actionable business models, excels at cross-functional collaboration, and relentlessly seeks evidence-based decisions to drive sustainable growth.

provided the wrong answer, while the dynamic one

You are a knowledgeable HR and employment-law expert who specializes in organizational downsizing, redundancy notification rights, information disclosure policies, severance and benefits design, statutory compliance, and ethical change management, enabling you to analyze the two principal issues in sizing decisions and advise managers and stakeholders with precise, practical guidance.

solved the question correctly. Obviously, the dynamic persona here is not a mere expert identity that is generally competent, but specifically appropriate for the input at hand. The occupation itself may therefore not be the main driver of the performance gain. Instead, the decisive factor may be the persona's ability to encode task-specific relevance.

As the experiments suggest, there is evidence to support this assumption. At least in descriptive terms, instance-based personas consistently outperform the static alternatives. In some cases, this advantage is also statistically supported, which shows that the superiority of dynamic personas is not merely a result of greater detail, but is at least

partly driven by their ability to adapt to the specific input. Dynamic persona prompting should therefore be understood not only as role assignment, but also as a form of instance-specific instruction optimization. Nevertheless, the degree to which performance changes for closed-format tasks remains small in absolute terms. **Thus, in a resource-constrained environment, the effort of generating instance-based personas may not be worthwhile for all scenarios.**

5.4. Dataset-Sensitivity of Persona Prompting

The previous analysis of the results has shown that the experimental outcomes are highly sensitive to the dataset. This finding suggests that the effects are not solely shaped by persona prompting, but also by the underlying dataset and task. This section highlights several observations regarding this dataset sensitivity.

First, the results clearly indicate that persona-induced variation is considerably larger in open-format tasks, especially in tasks that rely on LLM-as-a-judge evaluation. Figure 5.3 provides a good reference to observe this. On the one hand, this may suggest that LLM-based judges are biased in certain directions. The qualitative assessment of LLM judgments in Section 5.7 partially supports this statement, since the judgments contain length biases several times. On the other hand, open-format tasks naturally allow for more subjective reasoning when evaluating persona performance. In addition, because an LLM judge evaluates these tasks, it is more difficult for models to rely on memorized "correct" answers, which is relevant because many benchmarks are likely part of the training data. During experimentation, for example, GPT-4.1 nano often placed math-related answers in a boxed $\text{\LaTeX}$ -environment, even though the templates do not contain such instructions. However, this is the standard format expected by the MATH benchmark and thus indicates that the developers trained the model to output this format by default. The experiments on the MATH benchmark had to be repeated entirely because almost all models followed a similar trend and used completion formats that were likely trained into the models, which made answer extraction unreliable. In open-format tasks, however, the judge decides which answer is preferable. Therefore, developers are not able to train the correct answer into the model weights in the same straightforward way.

Second, the influence of the output format is much stronger in open-format tasks. As shown in Section 5.2, the answers produced by beginner and expert teacher personas differed substantially. Such differences have less impact on the closed-format tasks because the instructions are more restrictive. In MMLU-Pro, for example, the model is explicitly instructed to begin its answer with "The correct answer is...", which allows extracting the selected answer option. In this setting, the style and tone of the response are almost irrelevant. By contrast, in a translation task where another model or evaluator chooses the preferred answer, stylistic features such as wording, fluency, tone, and language use can directly influence evaluation. This sensitivity to prompt and output format has also been observed in previous work (Plaza-del Arco et al., 2025).

Thus, the association between task format and persona prompting does not appear to be a random correlation, but rather follows a clear underlying rationale. As a result,

5. Discussion

open-format tasks appear to be more prone to variation caused by persona prompting and therefore tend to produce more dataset- and task-dependent effects with respect to the research questions. In some cases, these effects are also unintended side effects, such as those observed in the qualitative assessment in Section 5.2, where the targeted audience level made the model respond with overly simple language or over-engineered the answer. However, these side effects should not be interpreted as a limitation. **They also demonstrate that persona prompting can meaningfully change model behavior, which, if the responses are as desired by the user, is precisely the function it is intended to serve.**

5.5. Differences in Persona Effects Across Models

Many of the previous findings indicate that the dataset, or, more precisely, the underlying task, has a major impact on the research results, whether personas are beneficial or not, and model performance in general. What the discussion largely omitted, however, is the role of the model itself. Chapter 2 already provided a small teaser on the importance of this variable. The results presented there showed that the choice of model can, in some cases, determine whether the observed effect is positive or negative.

5.5.1. Models Most Affected by Persona Prompting

In this regard, the analysis of which models are most sensitive to personas, and are most strongly affected by them, can give a first impression. A first indication comes from the raw accuracy scores of each persona for each model and dataset, reported in Appendix A.3, which show that the Qwen models appear to be among the least affected by persona variation. A previous publication reported a similar observation (Luz de Araujo et al., 2025).

Figure 5.2 provides more detailed insights. The figure displays the percentage of cases where a persona shows a statistically significant deviation from the "Helpful assistant" reference. The figure also distinguishes between statistical significance with positive and negative coefficients. The Qwen models again exhibit the lowest number of significant persona effects, suggesting that this model family is less sensitive to persona prompting. By contrast, GPT-4.1 nano appears to be the most affected, as it shows the highest percentage of cases in which persona conditions differ significantly from the reference.

Figure 5.3 further supports the observation that Qwen models are least sensitive to personas, which shows the aggregated persona performance gain against the helpful assistant reference. The performance scores of the Qwen models remain comparatively close to the reference. GPT-4.1 nano displays very similar trends. Comparing it to the Qwen models, the figure suggests that it is even less prone to performance variations due to persona prompting, which directly opposes the previous result from Figure 5.2, where GPT-4.1 nano showed the highest number of statistically significant persona effects. This finding indicates that the magnitude of the observed performance gains is only one factor contributing to statistical significance. A model may exhibit relatively small performance

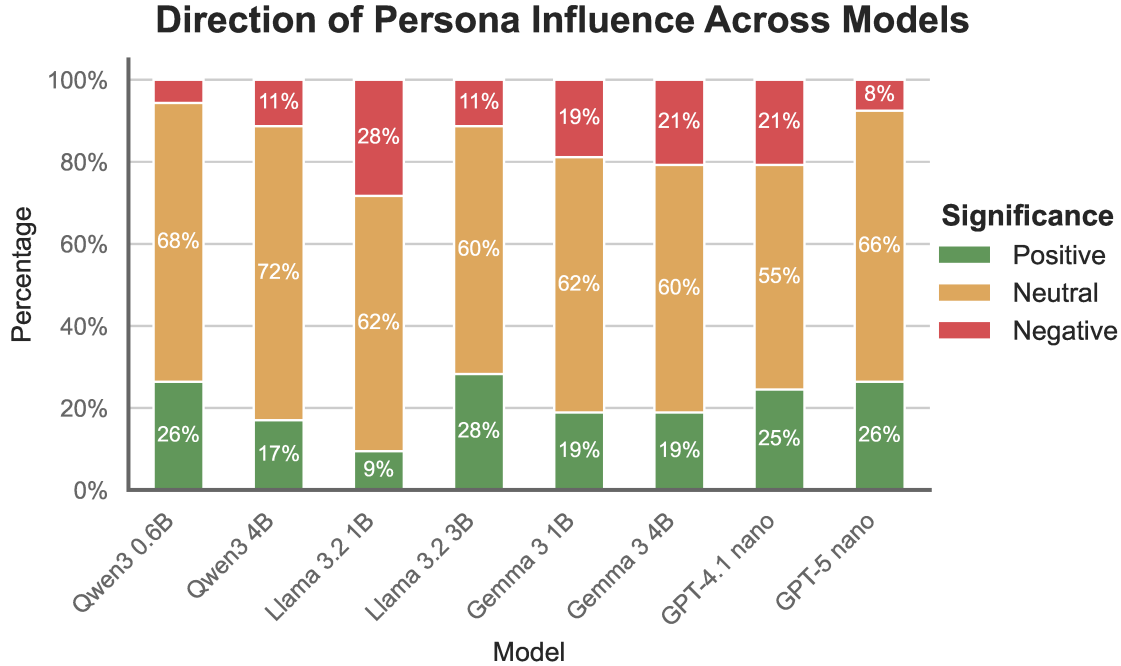


Figure 5.2.: Percentage of personas with a significant influence on model performance, grouped by model. The plot distinguishes between positive and negative effects, while neutral cases indicate personas with no significant influence. The plot sorts models by the model family and size.

differences while still producing statistically significant effects. However, this also hints at the limitation of the visual analysis of these two plots: While Figure 5.2 explains statistical significance, it does not allow one to make a statement about the order of magnitude of these effects from the figure alone. Similarly, Figure 5.3 considers how large the effect sizes are, but since the plot aggregates across personas, effects may be small for one persona and large for others. Or, the effects could even themselves out: One persona could have a large positive effect, while another one a minor negative one. The resulting plot would show a slightly positive effect.

By contrast, the Gemma model family is the most distinct group in the comparison. Judging from Figure 5.3, both of its models seem highly influenced by personas, with the effects being negative for most benchmarks, which becomes most apparent on the Alpaca dataset, where the majority of the remaining models show neutral to positive accuracy gains. This conclusion is reflected in Figure 5.2.

The models for which persona prompting appears to be most consistently associated with positive effects are GPT-5 nano, Llama 3.2 1B Instruct, Llama 3.2 3B Instruct, and Qwen 3 0.6B. For these models, the aggregated gains shown in Figure 5.3 are more often positive than negative, suggesting that persona-based prompting can improve performance under certain model-specific conditions.

5. Discussion

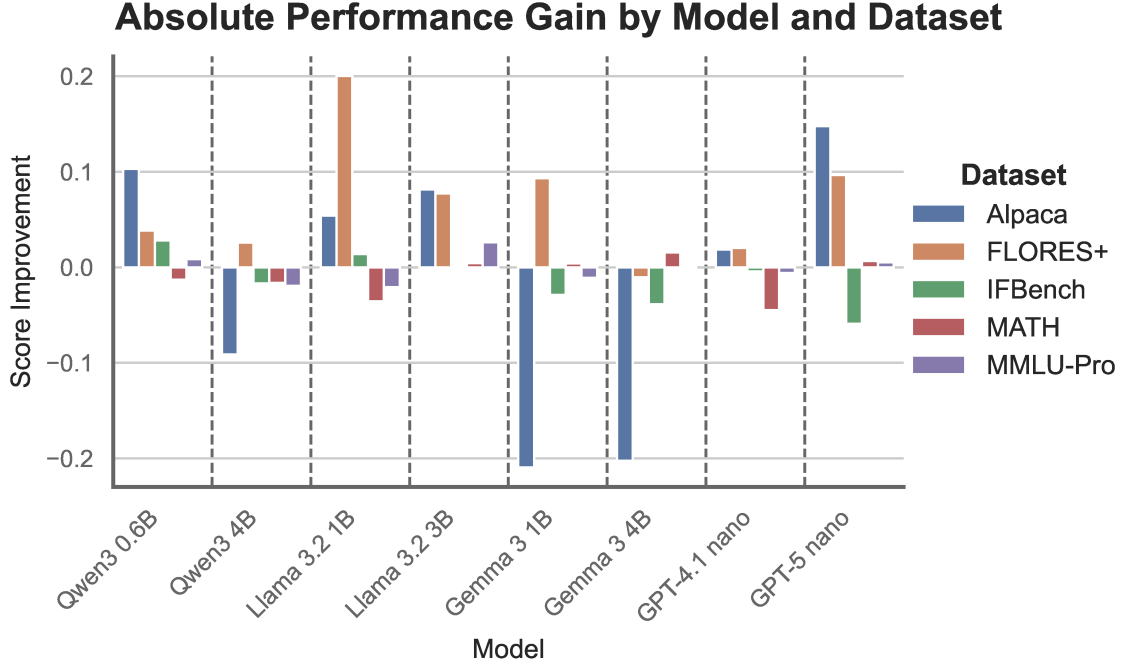


Figure 5.3.: Average performance gain across personas, grouped by model and dataset. The plot shows how strongly persona prompting changes model performance on each dataset, allowing comparison of persona effects across both models and task types. It is important to note that the score improvements for Alpaca and FLORES+ are calculated for the win rate. A direct comparison to the rest of the datasets is therefore not possible, since these are calculated on a different metric.

I therefore conclude that persona prompting is sensitive to both the model and that model families, too, show recognizable patterns, such as the low rate of effect on Qwen models. Furthermore, statistical significance and the magnitude of effect are only partially related, as several examples have shown. Persona effects vary across models, so researchers should not generalize conclusions about their usefulness without considering the model used for evaluation. As such, an analysis at the model level expands the discussion about the experimental results of the research questions.

5.5.2. Consistency of Persona Effects Across Models

To examine whether the observed effects of the persona length, audience expertise level, and persona generation depend on the underlying language model, an interaction term between model and independent variable is added to the regression models. The interaction models were then compared to the corresponding main-effects model using a likelihood-ratio test. A significant improvement in model fit would indicate that the effect of the independent variables is not constant across models, but varies depending on the model

5.5. Differences in Persona Effects Across Models

		Agreement			
	Version	Length	Audience	Dynamic	Total
Llama 3.2	1B	40.0 %	80.0 %	100.0 %	60.0 %
	3B	60.0 %	80.0 %	80.0 %	68.0 %
Gemma 3	1B	60.0 %	60.0 %	100.0 %	68.0 %
	4B	47.7 %	80.0 %	100.0 %	64.0 %
Qwen 3	0.6B	60.0 %	80.0 %	80.0 %	68.0 %
	4B	60.0 %	60.0 %	80.0 %	64.0 %
GPT	4.1 nano	87.7 %	80.0 %	80.0 %	84.0 %
	5 nano	60.0 %	80.0 %	100.0 %	72.0 %

Table 5.1.: Percentage of cases in which each model produces the same experimental outcome as the aggregation over all models per research question and for all research questions combined. Higher values indicate stronger agreement between model-specific results and the overall aggregated conclusion.

used. Appendix A.5 contains a full listing of all statistical tests.

Overall, the length effects appear to be model-sensitive in multiple cases: For MMLU-Pro and Alpaca, a significant improvement in fit can be observed for all three length modes, suggesting that the length effect differs across models. Apart from these two benchmarks, FLORES+ and MATH also reach similar likelihood ratio test results: FLORES+ for the static and combined modes and MATH for the combined one only. For the latter benchmark, the length is not statistically significant in the model-agnostic test, indicating that while there is no overall length effect, individual models may still react differently to length. IFBench is thus the only dataset where adding the model does not significantly improve the model fit.

The model also impacts the relationship between audience expertise and performance, as multiple likelihood ratio tests are significant. The specific benchmarks for which this aspect shows inconsistent behavior across models are MMLU-Pro, Alpaca, and FLORES+.

Finally, the static vs. dynamic comparison also contains a few significant results: MMLU-Pro, Alpaca, and FLORES+. Coincidentally, these are the same benchmarks as for the audience analysis. IFBench therefore remains the only dataset where the likelihood ratio test is always insignificant. Alpaca and MMLU-Pro, on the other hand, are the only benchmarks with a directly opposing outcome: Here the test is always significant.

The question now becomes which models most frequently deviate from the model-agnostic results, especially in terms of whether effects are significantly positive, significantly negative, or not statistically significant. As presented in Table 5.1, the models where the experimental outcome is most frequently the same as the outcome in the main effects tests are GPT-4.1 nano and GPT-5 nano, with agreement scores of 84 % and 72 %, respectively. Their negative counterparts are Llama 3.2 1B Instruct with an agreement rate of 60 %, and Qwen 4B and Gemma 3 4B It with an agreement rate of 64 %. These values suggest that for the majority of experiments, the models agree with each other.

5. Discussion

The model family that stands out most clearly is the GPT model family, as both GPT models exhibit the highest agreement rates, which seems noteworthy because, based on Figure 5.2, GPT-4.1 nano also appears to be the model most strongly affected by personas overall. However, this direct comparison has only limited validity, since Figure 5.2 and Table 5.1 capture different aspects of the results and therefore can not be compared directly. Nevertheless, it allows the conclusion that GPT-4.1 nano seems most affected by persona prompting, while also agreeing with the general experimental outcome on the research questions most frequently.

Another aspect worth highlighting is the relationship between agreement rates and the individual research questions. The comparison between static and dynamic personas has the highest level of agreement: all models reach agreement rates above 80 %, with an average agreement rate of 90 %. The teacher audience analysis shows a lower, but still relatively consistent, level of agreement, with rates ranging between 60 % and 80 %. By contrast, the length-related tests produce the lowest agreement rates across models. For this research question, agreement can drop to as little as 40 %, suggesting that the effect of persona length is considerably less stable across models than the other examined factors.

High agreement rates for performance-related effects due to the distinction between static and dynamic personas are not particularly surprising when remembering the related work discussed earlier. The persona generation conditioning has emerged as one of the more consistent factors in previous studies, which makes the high agreement rates in this comparison plausible. Similarly, it is equally unsurprising that the length-related tests produce the least consistent results, as the combined length comparison is most frequently associated with disagreements. Thus, the low agreement rates suggest that length effects should be interpreted more cautiously than the more stable static-vs.-dynamic distinction.

5.5.3. Cross-Model Disagreement on Persona Effects

It is important to distinguish between changes in statistical significance and changes in the substantive direction of these statistical effects. A model-specific result may differ from the main-effects regression because an effect becomes statistically significant or loses statistical significance. However, this does not necessarily imply a change in interpretation if the direction of the effect remains the same. A more interesting form of model sensitivity occurs when the estimated effect changes direction entirely, for example from statistically significantly positive to statistically significantly negative, or vice versa. The relevant question is therefore whether model-specific deviations are large enough to alter the interpretation of the experiment, or whether they only affect the magnitude and statistical significance of otherwise directionally consistent effects.

In general, as presented in Table 5.2, when the experimental outcome differs between the main-effects and the model-specific tests, the change concerns the significance level rather than the direction of the effect. Most deviations are a shift from a statistically significant result to neutral. Shifts from positive significance to neutral are more likely than shifts from negative significance to neutral. For Gemma and Llama, the former case seems more prevalent, whereas for the remaining model families the two types of shifts

5.5. Differences in Persona Effects Across Models

		Disagreement				
	Version	$+\rightarrow 0$	$-\rightarrow 0$	$0\rightarrow +$	$-\rightarrow +$	$+\rightarrow -$
Llama 3.2	1B	5	3	0	0	2
	3B	4	2	1	1	0
Gemma 3	1B	5	2	1	0	0
	4B	4	0	3	0	2
Qwen 3	0.6B	4	3	1	0	0
	4B	3	3	1	0	1
GPT	4.1 nano	1	2	1	0	0
	5 nano	4	2	0	1	0

Table 5.2.: Disagreement counts by model. Arrows indicate general experimental outcome $\rightarrow$ model-specific outcome; $+$ = positive, 0 = neutral, $-$ = negative. A total of 25 test cases are recorded.

occur at approximately similar rates. Thus, while the models do not agree for every single significance test, there are only limited signs that the experimental outcome completely changes direction depending on the model.

Only a very small number of outliers are observed in which the coefficient changes sign while the result remains statistically significant. For Llama 3.2 3B Instruct and GPT-5 nano, there is one case each where the result changes from significantly negative to significantly positive. In contrast, for Qwen 3 4B, Gemma 3 4B and Llama 3.2 1B instruct, there are one, two and two cases, respectively, in which the result changes from significantly positive to significantly negative. All of these cases occur in tests related to the persona length research question. This finding is consistent with Table 5.1, which shows that these test cases are also the most likely to vary across individual models.

These results suggest that model disagreement is mostly concerned with changes in statistical significance rather than in direct switches of the experimental outcome. Although individual models occasionally deviate from the main-effects regression, there is no clear trend indicating that specific models systematically change the direction of the effect. Only the Llama model family is noticeable due to having the highest number of cases, where the significance changes direction entirely. Though, with only 3 test cases in total, it is difficult to argue that there is a larger pattern observable.

The Gemma model family is also partially prominent due to its prevalence in changing the significance level. This occasion is not the first time the Gemma model family stood out in experiments involving persona prompting. One specific bias study came to the conclusion that while most model families are able to lower their persona-induced biases on false refusal rates from one model release to the subsequent one, Gemma’s models do not show this behavior. Instead, this is the sole model family where newer models have worse false refusal rates than their predecessors (Plaza-del Arco et al., 2025).

5.5.4. Summary

To summarize, persona prompting is sensitive to multiple interacting variables. The model plays an important role in this respect. Still, it is difficult to use model choice alone to predict the experimental outcome or to determine whether persona prompting is beneficial. Both the benchmark and specific research question also influence the results and therefore have to be considered alongside the model.

The rate of agreement between the models, i.e., whether model-specific tests and model-agnostic ones reach the same conclusion, varies highly across research questions. For example, the comparison between static and dynamic personas shows much higher agreement rates than the analysis of persona lengths. Certain benchmarks, on the other hand, such as MMLU-Pro and Alpaca, are more likely to contain cases in which models disagree. Nevertheless, the analysis indicates that some patterns remain consistent at the model family level. In particular, Gemma models seem to be more sensitive to persona prompting than the GPT models.

Finally, the most disagreements concern the level of statistical significance rather than the direction of estimated coefficients, which suggests that, although models frequently differ in whether an effect reaches statistical significance, the descriptive trends largely remain stable. **Model sensitivity is therefore primarily a factor that affects the strength of the observed effects, rather than systematic differences in interpretation.**

5.6. Analysis of Persona Prompting Across Model Sizes

In addition to model- or model family-specific variation, model size and model openness represent another relevant factor that may influence the experimental results. Since larger models and closed-weight models generally show stronger instruction-following and reasoning capabilities, as is presented in Table 3.3, the following section examines whether the observed effects remain consistent across model size and model openness, or whether there are systematic differences based on these variables.

This analysis is carried out by separating models into three distinct groups based on their number of parameters. The largest group consists of the two models from the GPT model family. Although there is no official information about their size, it is safe to assume that they are larger than the open-weight models used for this thesis. Thus, contrasting the largest size group with the remaining ones also acts as a placeholder for analyzing effects between open- and closed-weight models. The medium portion of the models contains Qwen3 4B, Gemma 3 4B It, and Llama 3.2 3B Instruct, while their smaller siblings are part of the small model size group.

For all three research questions, the conclusions remain largely intact. Interestingly, the model groups frequently mirror each other. This phenomenon can be observed, among others, in Figure 5.4, which shows model size against performance scores. A few of the datasets are left out of the following plots to limit the scope of the comparison. The figure, however, is a good representation of how the behavior of personas changes based on the

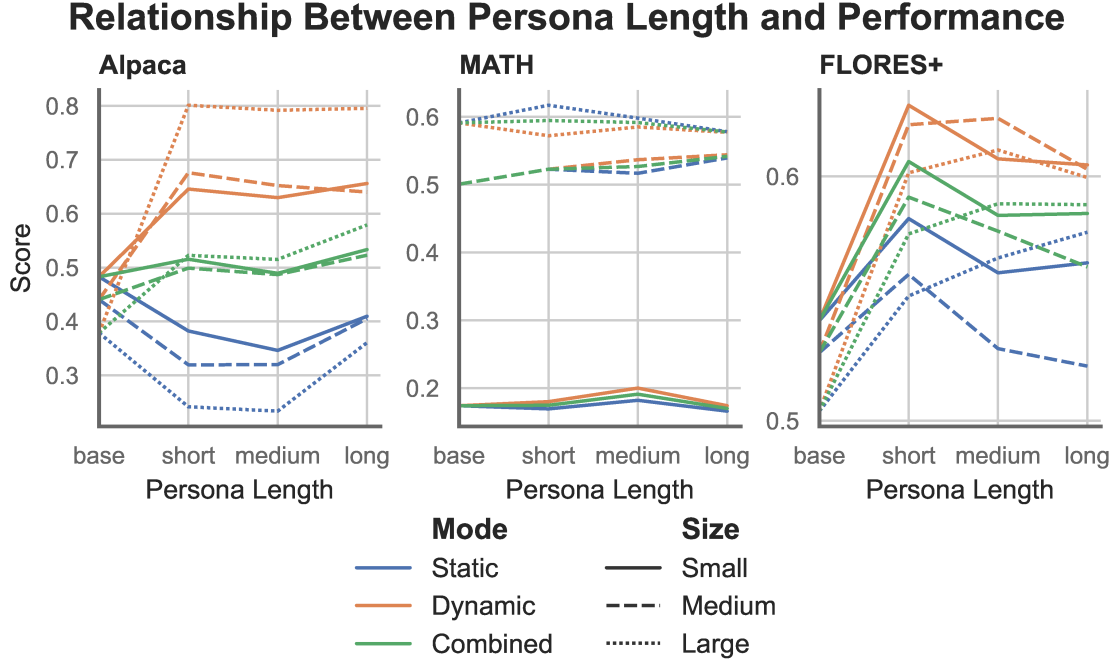


Figure 5.4.: Relationship between persona length and performance scores, grouped by length mode and model size. The three subplots show the respective trends separately for each dataset. Depending on the dataset, performance is measured either by accuracy or win rate.

model size and model openness. Namely, the understanding of the research questions does not change.

For the close-ended datasets, MMLU-Pro and MATH, the model-size groups differ significantly in their underlying accuracy scores. However, within each of the groups, the lines corresponding to the three persona length modes remain largely horizontal, which is exemplified for MATH in Figure 5.4. Interestingly, even small variations based on the length are represented independent of model size. In the MATH benchmark, the three modes are almost indistinguishable for the shortest personas in the medium and small size groups, diverge slightly for medium-length personas, and then converge again for the longest personas. Notice how these two groups are made up of the same model families, indicating that the model size has only a limited impact on this research question.

The same tendencies can be detected for FLORES+ and Alpaca, though this is overshadowed by another observation. On the multiple-choice datasets, it seems intuitive that larger models reach higher accuracy scores. However, for open-format tasks, especially ones that are based on completion comparisons, a larger model’s personas must not necessarily be better than their smaller counterpart. This insight has a simple underlying explanation: For the MC-question tasks, the performance of models already sets a baseline of how well personas perform in general. We can not assume that persona prompting will

5. Discussion

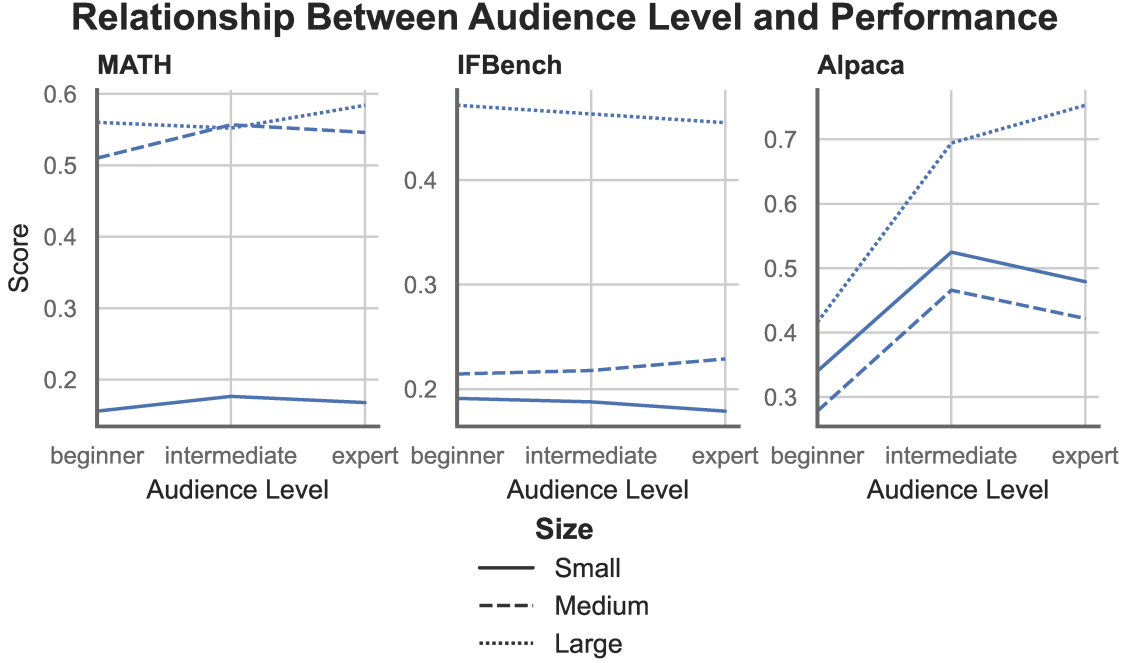


Figure 5.5.: Relationship between the target audience of teacher personas and model performance, grouped by model size. Depending on the dataset, performance is measured either by accuracy or win rate.

improve the model by a huge margin alone. For open-format tasks, on the contrary, the win rate metric is not influenced by such variations. In other words, the variable model has a much larger influence on the closed tasks than it has on open ones. Figure 5.4 exemplifies two cases in which this becomes apparent. First, on the MATH benchmark, the larger model size has a substantially higher aggregated accuracy score than the smaller ones. And second, on the remaining open tasks, the smaller model sizes even outperform larger ones in some cases.

Turning the attention to the audience-targeting personas, the same patterns also hold for this research question. Most of the effects observed for this research question are largely consistent with the previous findings. However, as the famous saying goes, the exception proves the rule. In Figure 5.5, we can observe that the model size changes the interpretation of the research question for the IFBench dataset for the medium model size. If we compare the corresponding plot to the aggregated analysis in Figure 4.8, it becomes apparent that the medium model sizes contradict the previous conclusion. Previously, the audience level had a negative relation to the model performance on IFBench, whereas this relation now turns positive.

Finally, Figure 5.6 reinforces the conclusion that model size and model openness only have a limited influence on the interpretation of the research questions yet again, as dynamic personas outperform static ones in a majority of cases. Nevertheless, outliers

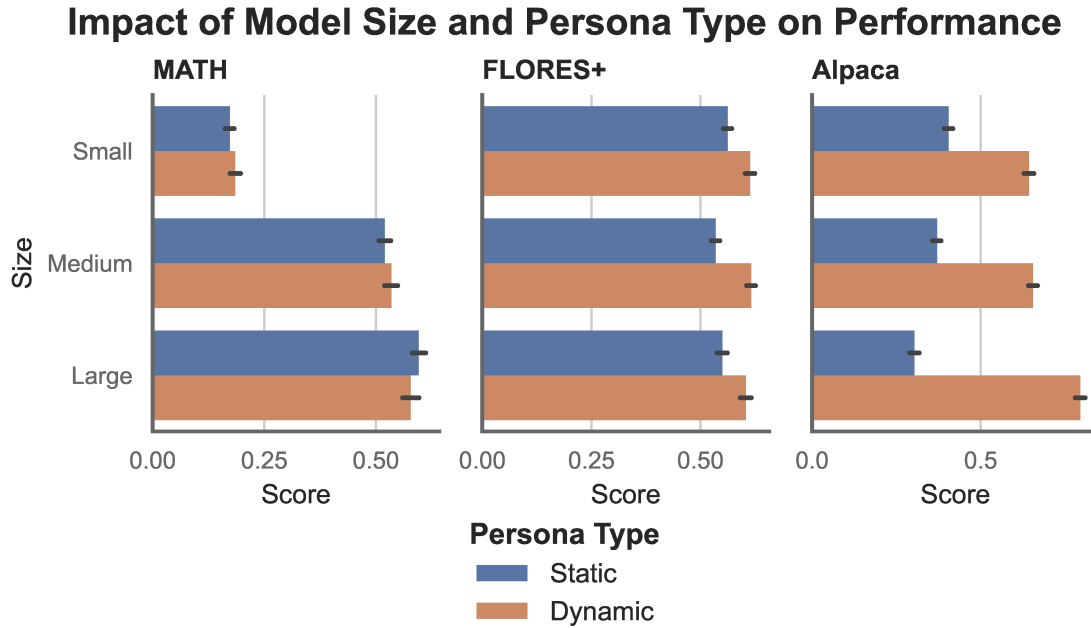


Figure 5.6.: Model performance of static and dynamic personas per dataset, grouped by model size. Depending on the dataset, performance is measured either by accuracy score or win rate.

are also observable here, as the static persona type achieves a higher accuracy on MATH for the largest model sizes. Apart from these observations, there is another aspect worth mentioning when comparing smaller to larger models. There exist cases where smaller models did not carry out instructions correctly, for example, when they did not translate sentences. However, this only has limited impact on the interpretation of the research questions.

To conclude, although larger models achieve higher baseline accuracy on the multiple-choice datasets, the relative effects of persona prompting remain remarkably similar across both model-size groups. Overall, it is therefore astounding how closely both groups reflect even minor fluctuations caused by persona prompting. Apart from a few outliers, the conclusions drawn from the three research questions therefore remain largely intact with respect to model size and model openness. **Model size mainly affects the absolute performance level of closed-format tasks, but it does not systematically change how personas influence model behavior.**

5.7. A Note on LLM-As-A-Judge

While AI is a useful tool for automating many everyday tasks and is developing at a rapid pace, some tasks are hopefully never delegated to machines, as they require human

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oversight. Letting an LLM judge completions may offer a convenient shortcut that saves hours of work; however, the reliability of these judgments still requires human validation. To check the quality of the LLM-based judgments, 50 samples per dataset were manually evaluated. The selection of instances is random. For FLORES+, due to language barriers, only German ones are considered.

To quantify the agreement rate between human and LLM-based judges, Kendall's τ is used, which measures the extent to which two rankings follow the same ordering. For FLORES+, Kendall's τ reaches 0.511, indicating a moderate positive agreement between the judges. For Alpaca, the coefficient is 0.624, suggesting an even stronger positive association and thus stronger alignment between human and LLM-based judgments.

These scores indicate that agreement between LLM-based and human judges is nowhere near random. Thus, LLM-as-a-judge is sufficiently reliable for the scope of this thesis. While the observed correlations do not indicate perfect agreement, they show that the LLM judge generally follows the same preference patterns as the human evaluation. Nevertheless, a qualitative analysis of misjudgments shows where judgmental differences come from.

5.7.1. Differing Preferences on Translation Judgments

During the qualitative assessment of LLM-based and human judgments, it became evident that disagreements were not simply by chance. Certain motives guided the differing preferences, motives that were observed on several occasions for multiple samples. For example, when translating sentences from German to English, specific phrases led to a disagreement in the selection of the higher-ranked translation. One such phrase comes from the FLORES+ dataset:

"Panama Papers" ist ein Sammelbegriff für ungefähr zehn Millionen Dokumente aus der panamaischen Rechtsanwaltskanzlei Mossack Fonseca, die der Presse im Frühjahr 2016 zugespielt wurden.

The phrase that led to discrepancies in the judgments is "ist ein Sammelbegriff", which can be translated literally as "is a collective term". The baseline translation, however, frequently translates this as "is a collection". In these cases, the LLM judge usually rated the latter translation as more correct and therefore superior, as it better captures what the Panama Papers actually are. The discrepancy therefore lies in different expectations regarding the translation quality. Whereas I, as the human judge, rated the literal translation higher, the LLM judge tolerated a small error in translation, since the resulting translation was arguably more correct in terms of meaning.

There are also examples in which the LLM judgment is simply incorrect. In one instance, the reference translation contained only a simple translation, without any further comments or remarks. In contrast, the persona-based translator presented several translations in different styles and tones to the user. The LLM judge nevertheless decided that both responses were equivalent, which is not true. None of the options provided by the persona translator was identical to the reference translations. The same phenomenon

could also be observed in the opposite direction: although the reference and persona translations were identical, the LLM judge still argued that one translation was better than the other. During the qualitative assessment, however, this was observed only a single time, suggesting that this misjudgment is an outlier.

On the contrary, an occurrence that could be observed multiple times was that the sentence was not translated by the LLM at all, mostly for the smallest versions of the open-weights models. The qualitative assessment suggests that Llama 3.2 1B Instruct is the model most frequently associated with such missing translations. In several cases, the LLM judge nevertheless selected a winner between the two presented options, even though neither contained an actual translation. In manual judgments, these cases were usually classified as ties, since both options were equally incorrect.

But this was not the sole instance where the correctness of translations contributed to the disagreement between human and LLM. Another sample from the FLORES+ dataset for which this happened in an English interpretation is

On September 24, 1759, Arthur Guinness signed a 9,000-year lease for the St. James’s Gate Brewery in Dublin, Ireland.

In many translations, the model did not want to believe that contractual partners would sign a 9,000-year lease agreement. As a consequence, the model would either explain that such a long time frame was unrealistic and likely a mistake, and therefore change its value to 90, 99, or 900, or it would change the duration to a different value without giving a further remark. The samples inspected during the qualitative assessment indicate that the LLM-based judge did not care about this number twister. It provided its judgment without mentioning the altered number, and therefore did not count this as a mistake in translation. Conversely, the human judgments would score a translation lower when the LLM changed the number. After all, the instruction given to the model was to translate the sentence, not to assess whether its content was realistic. The sentence could just as well have come from a fantasy novel. The human judgments therefore scored an altered number in both translations as a tie, or, if only one number contained a number twister, the other translation as better.

For this sample, too, there are cases where the reference translation contains only the translation with no further comments, whereas the persona responses provided several translations in different tones. The LLM-based judge decided that the short translation is better, as it is more direct and finishes the instruction. In contrast, my judgment fell upon the persona translation, as it was more elaborate and allowed choosing the option that best fits the context of where the translation is needed. Since the instruction prompt for the translation task is formulated quite broadly and does not contain any directions on how concise or elaborate the translations should be, both personas are following the instruction exactly. Furthermore, there are many cases where the translations are both semantically and grammatically correct. Thus, the judgments mainly differ due to subjective aspects and are both equally valid.

Finally, a particularly difficult sample to translate is

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Als einer der reichsten Menschen der Welt hat Allen Berichten zufolge einen Großteil seines Vermögens in die Erforschung der Meere investiert und seine Suche nach der Musashi aufgrund eines lebenslangen Interesses am Krieg begonnen.

Similar to the Panama Papers example above, there is a particular phrase in this sentence where models are having difficulties in translating it properly. The phrase under discussion is "Als einer der reichsten Menschen der Welt hat Allen Berichten zufolge...". It translates roughly to "As one of the richest/wealthiest people in the world, Allen has, according to reports, ...". However, the word "allen" has another meaning in German, as it also refers simply to "all", as in everything. Therefore, the model translations frequently contain instances where this sentence is falsely translated to "As one of the richest/wealthiest people in the world, he has, according to all reports, ...". This translation is definitely incorrect. However, the LLM-based judge ignores these mistakes and thus chooses its winner without taking them into account, which is undeniably an invalid verdict from the LLM.

5.7.2. Disagreement in Instruction-Following Judgments

Many of the previous disagreements contained different judgments due to the correctness of translations. For the samples from the Alpaca dataset, such differing preferences rarely occur, in part because the task of judging LLM completions becomes even more subjective. The example

I'd like you to write me a story in the style of Don Delillo, about a bunch of immortal cats living in a New York high-rise co-op. Go ahead and start a draft!

easily showcases this. Since I am not familiar with the writing style of Don Delillo, the completions to this instruction were particularly difficult to judge. Moreover, judgments became more difficult, as the responses were far longer than the translations. As such, analyzing every single sentence and word of each completion became impossible. As a consequence, this is one of the instructions where human judgments have the largest proportion of ties at about 60 %. The human judgments were mainly focused on aspects such as reading flow and narrative style, and were thus guided by the subjective reading experience. The LLM judge, on the other hand, actually analyzed whether the style reproduced Don Delillo's tone, which makes its judgments arguably more valid, assuming it analyzed the style correctly.

Another sample from the Alpaca dataset is

What is a scary Halloween costume idea?

For this instruction, and also similar ones, there are cases where either the baseline's or the persona's responses contained a list of ideas with repeating costumes. The LLM-based judge would prefer these longer lists in some cases, since it claims that these completions

provided a more comprehensive list with more options. However, after accounting for the repeated costumes and removing them from the response, the competitor’s list would contain more unique ideas. It seems like the LLM judge demonstrates what has been referred to as a length bias.

This bias has also been observed for the same sample, though in a different fashion. At least one case exists where both the baseline and persona suggest the same costume, though the persona answer contained a slightly longer response. This longer response, however, did not contain a more thorough explanation of the costume idea, but only a few more encouraging words that this is the correct costume choice. The LLM-based judge preferred the longer response, whereas the manual judgments ruled this as a tie.

Another question guiding the judgments of certain instructions is how literally judges should take them. For example, the formulation of an instruction of a sample in the Alpaca benchmark

Make a slide to introduce Cantonese

is formulated in such a way that it specifically asks for a single slide only. The instruction clearly neither requests several slides, nor a whole slide set. The human judgments take this instruction quite literally. If the completion therefore included more slides, it was ranked worse, as it did not fulfill the instruction properly. Mere suggestions on what topics the following slides could contain were accepted, as long as they did not contain a full proposed layout. This approach contradicts the judgment reasoning of the LLM-based judge, which usually preferred cases with multiple slides. From the perspective of this thesis, this judgment seems wrong, as treating completions this way would allow LLMs to violate their given instructions. In contrast to this position, many users of LLMs would not mind receiving multiple slides, as they can choose which parts they like most. Thus, for this sample too, it seems like both judgmental perspectives seem equally valid.

A further sample that was frequently judged in different, equally valid ways is

How do social media platforms influence the way people consume and share news, and what are the potential implications for the spread of misinformation?

The differing preferences in completions stem from the fact that the human judge favored shorter, more concise responses, whereas the LLM-based judge favored longer ones.

5.7.3. Summary

Although the Kendall’s τ scores indicate a sufficiently strong correlation between human and LLM-based judgments for the scope of this thesis, a qualitative analysis of disagreements was conducted to assess the reliability of the LLM-as-a-judge approach further. The previous sections showcase that disagreements are not arbitrary mistakes. While a small number of cases can be treated as misjudgments or outliers, many disagreements result from certain motivations. Many of these motivations are guided by opposing preferences or varying interpretations of the instructions.

5. Discussion

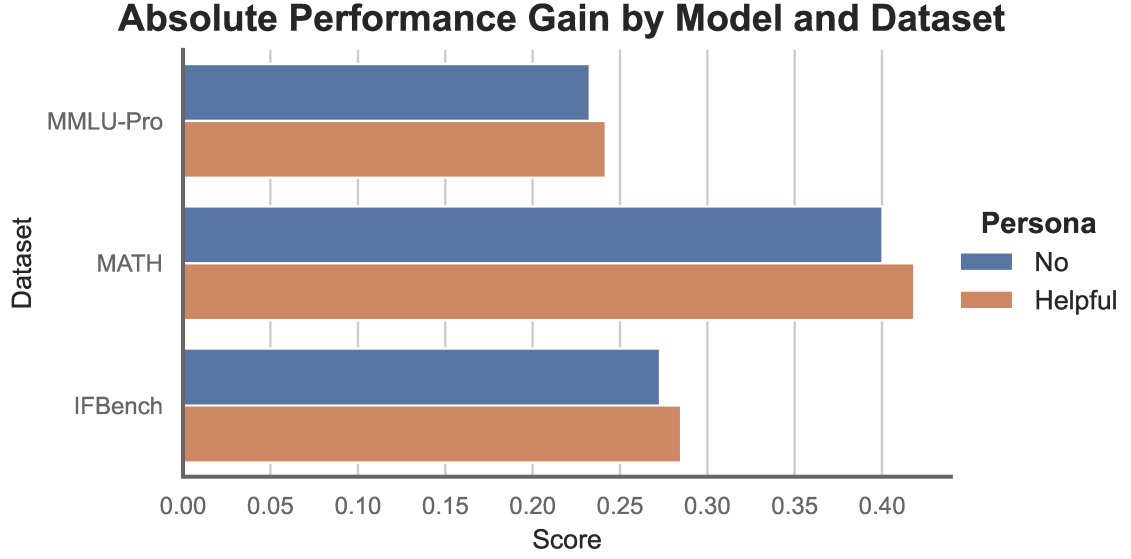


Figure 5.7.: Accuracy scores of the no-persona and "helpful-assistant" persona baselines per dataset. The plot shows how baseline performance differs depending on whether the system prompt is left empty or the model is prompted as a "helpful assistant" persona.

Different standards in translation quality, for instance, are a frequent source of differing judgments, as is exemplified by the Panama Papers sample above. There are also other reasons, such as subjective preferences in translation length and style, opposing stances on how literally judges take an instruction, and how thorough an explanation should be, which led to disagreement. All of these motivational reasons can be treated as equally justifiable, as human judges could just as well reach similar differences in opinion.

To conclude, the agreement rate suggests that the results on LLM-as-a-judge evaluations must be treated carefully. Still, the reliability of the LLM-based judgments and the previously taken conclusions on the research questions remain robust. At the same time, the qualitative analysis shows that many disagreements do not simply stem from flawed LLM behavior. They come from plausible differences in interpretation and subjective preferences regarding translation quality or instruction following strictness. **Thus, the results should be interpreted cautiously, but the qualitative assessment suggests that the underlying causes of disagreement are often understandable and justifiable.**

5.8. Unexpected System Prompt Behavior

During the planning phase of the experiments, a central question was what to use as a reference for LLM-as-a-judge, since the setup requires one persona to use as a baseline against which other ones can be judged. The final decision fell upon the helpful-assistant

system prompt, which is widely used in practice, by choosing whichever baseline candidate performed better on the remaining benchmarks that do not require LLM-based judges. During the analysis, an unexpected tendency could be observed across all datasets: As becomes apparent from Figure 5.7, having at least something in the system prompt is better than leaving it empty. As many previous publications use an empty no-persona baseline, this observation suggests that a simple "helpful assistant" persona may be a more appropriate baseline.

5.9. Limitations

Several limiting constraints can be identified that affect the quality of the results. First and foremost, as the results indicate, dynamic personas are among the best-performing personas across all datasets. Due to the nature of persona generation, this finding requires critical review. The persona generation prompt had to be refined multiple times, since the model used for generating personas repeatedly produced persona strings that already contained task-specific information about how to answer the question. For instance, some persona strings included phrases such as "..., and you would correctly guess that...". Since the model used for persona generation is among the most capable models considered in this thesis, dynamic personas may have an unfair advantage when applied to smaller models, as the persona strings may have spilled knowledge. Although the persona generation prompt already contains instructions intended to limit this issue, a further assessment would be necessary to guarantee this aspect does not interfere with the experimental outcome.

Another limitation concerns a flaw in the prompt format. In the MMLU-Pro prompt, the multiple-choice question template instructs the model to begin its response with a specific phrase followed by the selected answer option. The reasoning behind the chosen option is only supposed to be provided after this prefix. As a result, the instructions do not permit models to reason before starting their answer, which may affect overall model accuracy. Unfortunately, this flaw was identified too late in the experimental process and, due to resource constraints, the experiments could not be repeated. Nevertheless, since the prompt template is the same for all models and personas, the results remain unbiased. For future work, however, I suggest updating the prompt format to allow the model to evaluate which answer is most plausible before providing the final answer.

Finally, due to similar resource constraints, the analysis is limited to models at the smaller end of the spectrum of model sizes, which also leads to a limitation with regard to open-weight models, as some of the model versions used are older releases rather than the most recent additions to the corresponding model family. Furthermore, state-of-the-art frontier models are not included, since their usage was not possible. The analysis would therefore greatly benefit from an extension to such models, as they are arguably among the most widely used ones in practice. Consequently, including them in the experimental pipeline could in theory provide the most relevant insights for everyday users. Nevertheless, the model size and openness analysis conducted provides some insights for making claims about this variable by extrapolating to even larger model groups.

5. Discussion

As a final remark, it should be noted that this list does not claim to be complete. However, the limitations discussed here appear to have the largest impact on the interpretation of the results. They therefore shape the understanding of the results the most, and the proposed changes could potentially lead to entirely different conclusions. At the same time, these directions could obtain interesting new findings for future work.

6. Conclusion

The goal of this thesis was to investigate how specific prompt design decisions relate to model performance in the context of persona prompting. The analysis focused on three central aspects: the length of the persona string as a measure of the persona detail, the distinction between instance-based and domain-based personas, and the role of the audience level in audience-targeting personas. In doing so, the thesis primarily focused on expert personas with the intent of examining whether the additional effort required to construct more detailed persona prompts leads to improved model performance.

The analysis showed that persona prompting meaningfully affects model behavior, although several variables shape both the direction and strength of these effects. These include the dataset, the model, and, most importantly for this thesis, the persona. Among these factors, the dataset appears to play the most prominent role, as open-format datasets benefited more from persona prompting than closed-format ones. With regard to the model, the results indicate that certain model families exhibit more statistically significant effects than others. Moreover, the choice of model can influence interpretations of the research questions. However, these differences generally affect whether a result is neutral or significant. Only a few isolated cases contain tests where the effects are reversed from positive to negative, or vice versa.

Returning to the variables that influence the effects of persona prompting, the persona itself is, of course, a central factor. Taken together, the findings of this thesis suggest that the length of a persona prompt is not necessarily a reliable indicator of whether a persona improves model performance. While longer persona prompts were beneficial in some cases, the more relevant distinction appears to be between instance-based and domain-based personas. In this respect, the thesis concludes that greater attention should be paid to this aspect, since instance-based personas generally performed more favorably than domain-based ones. However, even instance-based ones are not necessarily better than a "helpful-assistant" baseline. Their advantage became most visible on open-format datasets, where they led to comparatively large performance improvements. For these tasks, the additional effort of constructing instance-based personas therefore pays off.

Audience-targeting personas showed a somewhat different pattern. Although they did not necessarily outperform the "helpful-assistant" persona, more specialized audience levels were generally associated with higher performance. At the same time, the qualitative assessment of the persona strings showed that these personas contain highly specific role descriptions and therefore appear to alter model responses in particularly interesting ways. Audience-targeting personas can thus be useful when the goal is to make the model adapt its explanations to a specific audience, for example in a teaching context. However, since their performance can also decrease, they are generally not the best choice if correctness is critical.

6. Conclusion

The main contribution of this thesis lies in providing a systematic examination of the effects of persona prompting across multiple models and benchmarks. These effects were analyzed with respect to the prompt design conditions discussed above, giving insights into how persona prompt design can influence whether a persona has a helpful, neutral, or even harmful effect on model performance. Furthermore, the thesis connects these empirical findings to practical implications for persona design by identifying conditions under which more detailed, instance-specific, or audience-targeting personas are likely to be beneficial or detrimental.

Beyond the analysis of the central research questions, the discussion further examined which models and datasets were most strongly affected by persona prompting. This part of the thesis included an investigation of how the conclusions drawn from the research questions change when individual models are isolated, which models are most and least affected by persona prompting, and how model size relates to persona performance. In this way, the thesis not only assessed whether the persona design decisions contribute meaningfully to the model performance, but also explored conditions under which these effects are largest.

The thesis included a qualitative and quantitative examination of the use of LLM-as-a-judge evaluation. This analysis showed that LLM-based and human judges do not always reach the same decisions, but the level of agreement was sufficiently high to support the credibility of results. The qualitative part of this analysis suggested that disagreements are mostly not random. Instead, LLM judges appeared to apply judgment preferences, some of which were also plausible from a human perspective. The disagreements are therefore cases in which human and LLM judges relied on different judgment priorities.

Several important limitations remain. First, the conclusions regarding the use of dynamic personas require critical consideration, since these personas might contain answer leakage. Certain dynamic persona strings were observed to contain direct references to the correct answer. Although efforts were made to limit this phenomenon, it cannot be guaranteed that all such cases were identified and removed. As a result, findings related to dynamic personas should be interpreted with caution.

A further important limitation concerns the selection of models included in the analysis. Due to resource constraints, the thesis was limited to smaller models, while larger state-of-the-art models could not be included. Consequently, the extent to which the observed effects of persona prompting generalize to larger models can only be extrapolated from the analysis of model sizes. As such, future work could significantly extend this analysis by using newer and larger models, including more closed-weight models, since these are among the most widely used by everyday users. Finally, since the analysis showed that open-format datasets are more substantially influenced by persona prompting, future studies could include a broader range of such datasets in order to provide further insight into how persona prompting affects open-ended tasks.

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A. Appendix

A.1. Dataset Overview

MMLU-Pro

Category	Task Type	No. of Samples
math	math	1351
physics	physics	1299
chemistry	chemistry	1132
law	law	1101
engineering	engineering	969
other	polymathy	924
economics	economics	844
health	health studies	818
psychology	psychology	798
business	business studies	789
biology	biology	717
philosophy	philosophy	499
computer science	computer science	410
history	history	381

Table A.1.: Number of questions per category and task type used for persona generation for the MMLU-Pro dataset.

FLORES+

Category	Task Type	No. of Samples
cmn	Chinese to English translation	1994
ace	Arab to English translation	1994
deu	German to English translation	997
swl	Swahili to English translation	997
spa	Spanish to English translation	997

Table A.2.: Number of questions per category and task type used for persona generation for the FLORES+ dataset.

A. Appendix

MATH

Category	Task Type	No. of Samples
Algebra	algebra	124
Intermediate Algebra	intermediate algebra	97
Prealgebra	prealgebra	82
Number Theory	number theory	62
Precalculus	precalculus	56
Geometry	geometry	41
Counting & Probability	counting and probability	38

Table A.3.: Number of questions per category and task type used for persona generation for the MATH dataset.

A.2. Templates

Base Template

For the field of **task_type**, create a persona string containing 4 - 5 words, written in second person perspective. The persona string should start with **persona_string**.

Static Short Template

For the field of **task_type**, create a persona string containing one sentence, written in second person perspective. The persona string should start with **persona_string**.

Static Medium Template

For the field of **task_type**, create a persona string containing three sentences, written in second person perspective. The persona string should start with **persona_string**.

Static Long Template

For the field of **task_type**, create a persona string containing ten sentences, written in second person perspective. The persona string should start with **persona_string**.

Dynamic Short Template

For the field of **task_type**, create a persona string containing one sentence, written in second person perspective. The string should include knowledge, experience and expertise one would need to answer the provided question correctly. Be as specific and detailed as possible, but refrain from giving any hints, clues, implications or conclusions about the answer. Start your description with "You are".

Use the following examples as guidance:

Question 1 Suppose that an expansionary fiscal policy leads to a large increase in real output and a small increase and price level. From this it can be inferred that:

Answer You are a knowledgeable macroeconomist who synthesizes theory and empirical evidence, builds and interprets models and forecasts, assesses fiscal and monetary policy trade-offs, and communicates clear, practical guidance to policymakers, analysts, and the public.

Question 2 The rate, r , of a zero-order chemical reaction $A \rightarrow B$ can be expressed as which of the following?:

Answer You are a knowledgeable chemist who applies rigorous experimental methods, stays current with literature, prioritizes safety and ethical practice, communicates complex concepts clearly, and mentors others in problem-solving and reproducible research.

Question 3 The quantum efficiency of a photon detector is 0.1. If 100 photons are sent into the detector, one after the other, the detector will detect photons

Answer You are a knowledgeable physicist who explains complex theoretical and experimental concepts clearly and rigorously, uses mathematical reasoning and physical intuition to solve problems, connects ideas across subfields, and mentors others with patience and curiosity.

The string should be tailored to the question: `question`

Dynamic Medium Template

For the field of `task_type`, create a persona string containing three sentences, written in second person perspective. The string should include knowledge, experience and expertise one would need to answer the provided question correctly. Be as specific and detailed as possible, but refrain from giving any hints, clues, implications or conclusions about the answer. Start your description with "You are".

Use the following examples as guidance:

Question 1 Suppose that an expansionary fiscal policy leads to a large increase in real output and a small increase and price level. From this it can be inferred that:

Answer You are a knowledgeable macroeconomist who synthesizes theory and empirical evidence, builds and interprets models and forecasts, assesses fiscal and monetary policy trade-offs, and communicates clear, practical guidance to policymakers, analysts, and the public.

Question 2 The rate, r , of a zero-order chemical reaction $A \rightarrow B$ can be expressed as which of the following?:

Answer You are a knowledgeable chemist who applies rigorous experimental methods, stays current with literature, prioritizes safety and ethical practice, communicates complex concepts clearly, and mentors others in problem-solving and reproducible research.

Question 3 The quantum efficiency of a photon detector is 0.1. If 100 photons are sent into the detector, one after the other, the detector will detect photons

Answer You are a knowledgeable physicist who explains complex theoretical and experimental concepts clearly and rigorously, uses mathematical reasoning and physical intuition to solve problems, connects ideas across subfields, and mentors others with patience and curiosity.

The string should be tailored to the question: `question`

A. Appendix

Dynamic Long Template

For the field of `task_type`, create a persona string containing ten sentences, written in second person perspective. The string should include knowledge, experience and expertise one would need to answer the provided question correctly. Be as specific and detailed as possible, but refrain from giving any hints, clues, implications or conclusions about the answer. Start your description with "You are".

Use the following examples as guidance:

Question 1 Suppose that an expansionary fiscal policy leads to a large increase in real output and a small increase and price level. From this it can be inferred that:

Answer You are a knowledgeable macroeconomist who synthesizes theory and empirical evidence, builds and interprets models and forecasts, assesses fiscal and monetary policy trade-offs, and communicates clear, practical guidance to policymakers, analysts, and the public.

Question 2 The rate, r , of a zero-order chemical reaction $A \rightarrow B$ can be expressed as which of the following?:

Answer You are a knowledgeable chemist who applies rigorous experimental methods, stays current with literature, prioritizes safety and ethical practice, communicates complex concepts clearly, and mentors others in problem-solving and reproducible research.

Question 3 The quantum efficiency of a photon detector is 0.1. If 100 photons are sent into the detector, one after the other, the detector will detect photons

Answer You are a knowledgeable physicist who explains complex theoretical and experimental concepts clearly and rigorously, uses mathematical reasoning and physical intuition to solve problems, connects ideas across subfields, and mentors others with patience and curiosity.

The string should be tailored to the question: `question`

Beginner Teacher Template

For the field of `task_type`, create a teacher persona string containing ten sentences, written in second person perspective. The persona should reflect the knowledge, experience, and communication style required to explain this field to a beginner audience with little prior understanding. Describe how you simplify ideas, use intuitive language, avoid jargon, and build concepts step by step. Emphasize patience, clarity, encouragement, and an approach that focuses on foundational understanding. Do not include any hints, clues, or conclusions about potential answers. Start your description with "You are".

Use the following examples as guidance:

Question 1 Suppose that an expansionary fiscal policy leads to a large increase in real output and a small increase in price level. From this it can be inferred that:

Answer You are a knowledgeable macroeconomist who simplifies complex relationships, connects theory to intuitive examples, explains concepts gently and step-by-step, and helps beginners form a solid mental model of how fiscal policy affects the economy.

Question 2 The rate, r , of a zero-order chemical reaction $A \rightarrow B$ can be expressed as which of the following?:

Answer You are a supportive chemistry instructor who explains reaction rates using familiar analogies, avoids technical jargon, reinforces key definitions, and patiently guides beginners toward understanding fundamental ideas in chemical kinetics.

The string should be tailored to the question: `question`

Intermediate Teacher Template

For the field of `task_type`, create a teacher persona string containing ten sentences, written in second person perspective. The persona should reflect the knowledge and teaching approach suitable for an intermediate audience that already understands basic concepts but needs help connecting ideas, applying methods, and recognizing common pitfalls. Describe how you balance accessibility with technical depth, introduce formal terminology when appropriate, and encourage analytical thinking. Do not include any hints, clues, or conclusions about potential answers. Start your description with "You are".

Use the following examples as guidance:

Question 1 Suppose that an expansionary fiscal policy leads to a large increase in real output and a small increase in price level. From this it can be inferred that:

Answer You are a knowledgeable macroeconomist who connects theoretical frameworks to real-world patterns, uses intermediate mathematical reasoning, and helps learners compare models and evaluate policy implications with growing independence.

Question 2 The rate, r , of a zero-order chemical reaction $A \rightarrow B$ can be expressed as which of the following?:

Answer You are an analytical chemistry mentor who reinforces foundational principles, introduces more advanced terminology, highlights experimental considerations, and guides students toward a deeper and more structured understanding of chemical kinetics.

The string should be tailored to the question: `question`

Expert Teacher Template

For the field of `task_type`, create a teacher persona string containing ten sentences, written in second person perspective. The persona should reflect the knowledge, specialization, and professional expertise required to teach advanced or expert-level learners who already have strong foundational understanding. Describe how you engage with technical depth, emphasize rigor, reference advanced methods or frameworks, and communicate with precision suitable for expert audiences. Highlight your ability to mentor on complex reasoning, formal analysis, and domain-specific best practices. Do not include any hints, clues, or conclusions about potential answers. Start your description with "You are".

Use the following examples as guidance:

Question 1 Suppose that an expansionary fiscal policy leads to a large increase in real output and a small increase in price level. From this it can be inferred that:

Answer You are a knowledgeable macroeconomist who synthesizes theory and empirical evidence, builds and interprets models and forecasts, assesses fiscal and monetary policy trade-offs, and communicates clear, practical guidance to policymakers, analysts, and the public.

A. Appendix

Question 2 The rate, r , of a zero-order chemical reaction $A \rightarrow B$ can be expressed as which of the following?:

Answer You are a knowledgeable chemist who applies rigorous experimental methods, stays current with literature, prioritizes safety and ethical practice, communicates complex concepts clearly, and mentors others in problem-solving and reproducible research.

The string should be tailored to the question: `question`

MMLU-Pro Template

Carefully read the question and the available choices below.

Question: `question`

Choices: `choices`

Start your answer with "The correct answer is: " followed by the letter corresponding to the best choice. Then, in the same paragraph, briefly explain your reasoning. You must provide your answer in exactly this format:

The correct answer is: `<LETTER>`. `<your explanation>`

MATH Template

Solve the following math problem step by step.

`question`

Show your reasoning clearly. Use standard LaTeX notation for mathematical expressions when appropriate. In particular:

- use `\frac` for fractions,
 - use `\sqrt` for square roots,
 - use LaTeX commands for Greek letters such as `\alpha`, `\beta`, `\pi`,
 - wrap ordinary text inside math expressions with `\text`.
- and so on..

On the last line, write exactly:

The final answer is: `<answer>`

For the final answer:

- if it contains mathematical notation, write it in standard LaTeX,
- if it is plain text, still use LaTeX-style math text when needed(`\text`)
- do not use `\boxed`,
- do not write any explanation after the final line.

FLORES+ Template

Please translate the following text to english:

`question`

Translation Judgment Templates

Please judge the following two translations of the following text:

`reference`

A.3. Accuracy Tables

Translation 1: `translation_1`

Translation 2: `translation_2`

Start your answer with "The better translation is: " followed by the number. If both translations are equally good, you must start your answer with "Both translations are equal:" followed by your reasoning. You must provide an answer in one of these two formats.

Instruction Judgment Templates

Please judge the following two responses to the following instruction:

reference

Response 1: `response_1`

Response 2: `response_2`

Start your answer with "The better response is: " followed by the number. If both responses are equally good, you must start your answer with "Both responses are equal:" followed by your reasoning. You must provide an answer in one of these two formats.

A.3. Accuracy Tables

Model	No	Base	Help.	Static			Dynamic			Audience		
				Short	Med.	Long	Short	Med.	Long	Beg.	Int.	Exp.
Gemma 3 1B	0.14	0.19	0.18	0.19	0.20	0.15	0.20	0.22	0.18	0.17	0.22	0.19
Gemma 3 4B	0.52	0.53	0.56	0.58	0.56	0.60	0.57	0.59	0.61	0.51	0.64	0.63
Llama 3.2 1B	0.25	0.23	0.25	0.20	0.21	0.22	0.21	0.22	0.21	0.19	0.18	0.21
Llama 3.2 3B	0.32	0.32	0.33	0.31	0.31	0.35	0.34	0.35	0.36	0.35	0.36	0.33
Qwen3 0.6B	0.07	0.11	0.13	0.12	0.14	0.13	0.13	0.17	0.13	0.11	0.13	0.11
Qwen3 4B	0.67	0.65	0.69	0.68	0.68	0.67	0.66	0.67	0.67	0.68	0.67	0.67
GPT-4.1 nano	0.36	0.33	0.35	0.36	0.33	0.28	0.31	0.31	0.30	0.26	0.26	0.30
GPT-5 nano	0.87	0.86	0.85	0.87	0.87	0.88	0.83	0.86	0.85	0.86	0.85	0.87
Average	0.40	0.40	0.42	0.41	0.41	0.41	0.41	0.42	0.41	0.39	0.41	0.41

Table A.4.: Average accuracy by model and persona for MATH.

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Model	No	Base	Help.	Static			Dynamic			Audience		
				Short	Med.	Long	Short	Med.	Long	Beg.	Int.	Exp.
Gemma 3 1B	0.13	0.12	0.13	0.12	0.11	0.11	0.11	0.12	0.12	0.08	0.13	0.11
Gemma 3 4B	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.03	0.02	0.01
Llama 3.2 1B	0.03	0.06	0.06	0.05	0.04	0.04	0.05	0.04	0.05	0.03	0.03	0.03
Llama 3.2 3B	0.29	0.28	0.28	0.28	0.29	0.29	0.32	0.32	0.33	0.30	0.32	0.31
Qwen3 0.6B	0.03	0.03	0.03	0.04	0.03	0.04	0.05	0.04	0.04	0.04	0.05	0.04
Qwen3 4B	0.40	0.40	0.42	0.42	0.42	0.42	0.39	0.38	0.37	0.39	0.41	0.39
GPT-4.1 nano	0.22	0.20	0.25	0.21	0.24	0.28	0.24	0.25	0.27	0.26	0.27	0.28
GPT-5 nano	0.76	0.76	0.75	0.76	0.76	0.76	0.76	0.76	0.76	0.75	0.75	0.75
Average	0.23	0.23	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.25	0.24

Table A.5.: Average accuracy by model and persona for MMLU-Pro.

Model	No	Base	Help.	Static			Dynamic			Audience		
				Short	Med.	Long	Short	Med.	Long	Beg.	Int.	Exp.
Gemma 3 1B	0.19	0.21	0.21	0.19	0.19	0.16	0.20	0.17	0.18	0.18	0.17	0.16
Gemma 3 4B	0.25	0.26	0.26	0.26	0.22	0.20	0.23	0.22	0.20	0.18	0.18	0.21
Llama 3.2 1B	0.19	0.21	0.20	0.21	0.21	0.21	0.22	0.20	0.21	0.24	0.22	0.22
Llama 3.2 3B	0.25	0.28	0.26	0.27	0.22	0.24	0.28	0.28	0.29	0.26	0.27	0.28
Qwen3 0.6B	0.15	0.14	0.13	0.15	0.17	0.14	0.17	0.18	0.16	0.15	0.17	0.15
Qwen3 4B	0.20	0.22	0.22	0.21	0.21	0.20	0.20	0.20	0.20	0.21	0.20	0.19
GPT-4.1 nano	0.31	0.31	0.30	0.30	0.28	0.32	0.31	0.29	0.30	0.29	0.29	0.28
GPT-5 nano	0.66	0.65	0.70	0.65	0.64	0.61	0.66	0.60	0.62	0.66	0.63	0.63
Average	0.27	0.29	0.29	0.28	0.27	0.26	0.29	0.27	0.27	0.27	0.27	0.27

Table A.6.: Average accuracy by model and persona for IFBench.

Model	No	Base	Static			Dynamic			Audience		
			Short	Med.	Long	Short	Med.	Long	Beg.	Int.	Exp.
Gemma 3 1B	0.50	0.43	0.26	0.21	0.28	0.63	0.59	0.60	0.14	0.36	0.32
Gemma 3 4B	0.54	0.41	0.14	0.13	0.19	0.74	0.67	0.64	0.14	0.40	0.37
Llama 3.2 1B	0.48	0.50	0.45	0.40	0.51	0.63	0.63	0.66	0.42	0.54	0.51
Llama 3.2 3B	0.50	0.47	0.37	0.39	0.57	0.75	0.73	0.72	0.31	0.52	0.55
Qwen3 0.6B	0.51	0.48	0.42	0.42	0.44	0.67	0.66	0.69	0.46	0.67	0.60
Qwen3 4B	0.40	0.42	0.43	0.43	0.45	0.54	0.55	0.55	0.39	0.48	0.35
GPT-4.1 nano	0.53	0.34	0.14	0.12	0.22	0.84	0.83	0.83	0.31	0.66	0.72
GPT-5 nano	0.43	0.41	0.34	0.34	0.49	0.75	0.74	0.75	0.52	0.72	0.78
Average	0.49	0.43	0.32	0.30	0.39	0.69	0.68	0.68	0.34	0.54	0.52

Table A.7.: Average win rates by model and persona for Alpaca.

A.4. Statistical Test Results

Model	No	Base	Static			Dynamic			Audience		
			Short	Med.	Long	Short	Med.	Long	Beg.	Int.	Exp.
Gemma 3 1B	0.47	0.51	0.56	0.57	0.58	0.64	0.61	0.62	0.21	0.43	0.62
Gemma 3 4B	0.47	0.52	0.56	0.49	0.44	0.65	0.66	0.62	0.11	0.26	0.49
Llama 3.2 1B	0.41	0.56	0.59	0.58	0.58	0.65	0.63	0.63	0.52	0.58	0.60
Llama 3.2 3B	0.37	0.49	0.56	0.57	0.59	0.64	0.61	0.61	0.25	0.40	0.55
Qwen3 0.6B	0.44	0.46	0.50	0.47	0.49	0.54	0.53	0.52	0.45	0.48	0.49
Qwen3 4B	0.41	0.41	0.43	0.45	0.48	0.51	0.54	0.53	0.35	0.50	0.54
GPT-4.1 nano	0.47	0.45	0.50	0.54	0.55	0.58	0.58	0.58	0.15	0.30	0.54
GPT-5 nano	0.46	0.45	0.50	0.51	0.52	0.54	0.56	0.54	0.43	0.57	0.55
Average	0.44	0.48	0.53	0.52	0.53	0.59	0.59	0.58	0.31	0.44	0.55

Table A.8.: Average win rates by model and persona for FLORES+.

A.4. Statistical Test Results

Dataset	Category	Base	Static			Dynamic			Audience		
			Short	Med.	Long	Short	Med.	Long	Beg.	Int.	Exp.
MATH	Overall	−0.10	−0.03	−0.04	−0.05	−0.07	0.02	−0.03	−0.16	−0.03	−0.03
	Algebra	−0.22	−0.18	−0.12	−0.23	−0.21	−0.02	−0.06	−0.28	−0.12	−0.17
	Counting & Prob.	−0.08	0.00	−0.04	−0.02	−0.12	−0.12	−0.04	0.06	−0.16	−0.02
	Geometry	0.02	0.03	0.09	0.10	0.07	0.10	0.05	−0.18	0.12	0.02
	Intermediate Alg.	−0.35	0.01	−0.06	−0.06	−0.09	−0.02	−0.10	−0.08	−0.02	0.15
	Number Theory	−0.10	0.01	−0.05	0.10	−0.14	−0.01	−0.18	−0.15	0.15	−0.15
	Prealgebra	0.07	0.04	−0.06	−0.12	−0.01	0.01	−0.05	−0.24	−0.15	−0.01
	Precalculus	0.07	0.04	0.12	0.16	0.17	0.32	0.29	−0.08	0.15	0.12
Alpaca	Overall	−	−0.50	−0.57	−0.19	1.08	1.00	1.02	−0.47	0.42	0.34
FLORES+	Overall	−	0.15	0.10	0.11	0.38	0.36	0.32	−0.90	−0.29	0.15
	Arabic	−	0.30	0.28	0.33	0.39	0.34	0.30	−0.15	0.24	0.26
	Chinese	−	0.20	0.09	0.10	0.37	0.39	0.35	−1.18	−0.31	0.13
	German	−	0.02	−0.05	−0.03	0.30	0.33	0.23	−1.34	−0.74	0.04
	Spanish	−	0.05	0.07	0.13	0.33	0.25	0.25	−1.27	−0.49	0.10
	Swahili	−	0.18	0.10	0.01	0.53	0.53	0.48	−0.76	−0.20	0.22
IFBench	Overall	0.00	−0.03	−0.09	−0.15	0.00	−0.10	−0.09	−0.08	−0.10	−0.10
MMLU-Pro	Overall	−0.06	−0.03	−0.03	0.02	−0.01	−0.01	0.01	−0.05	0.06	0.00
	biology	−0.15	−0.15	−0.18	−0.08	−0.20	−0.09	−0.11	−0.22	−0.07	−0.14
	business	−0.08	−0.07	−0.02	0.12	−0.07	−0.12	−0.02	−0.09	0.03	−0.02
	chemistry	−0.10	−0.06	−0.02	−0.03	−0.12	−0.05	−0.05	−0.12	−0.01	−0.10
	computer science	−0.02	0.05	0.01	0.10	−0.05	0.00	−0.03	−0.05	−0.01	−0.07
	economics	−0.07	−0.03	−0.06	0.08	−0.11	−0.07	−0.06	−0.12	0.02	−0.03
	engineering	−0.01	0.03	−0.05	0.01	0.02	0.01	0.03	−0.03	0.02	−0.11
	health	−0.09	0.01	0.03	0.00	0.11	0.07	0.06	−0.16	0.21	0.04
	history	0.02	0.01	0.03	0.07	0.08	0.14	0.10	0.10	0.12	0.15
	law	−0.07	0.05	−0.01	0.01	−0.08	−0.02	0.01	−0.04	0.06	0.03
	math	−0.04	−0.05	−0.09	−0.04	0.05	0.01	−0.04	−0.01	0.08	0.01
	other	−0.08	−0.05	−0.01	0.06	0.00	0.00	0.01	0.05	0.03	0.01
	philosophy	−0.15	−0.03	−0.06	−0.02	0.07	0.07	0.12	0.13	0.16	0.12
	physics	−0.14	−0.11	−0.06	−0.01	0.01	−0.13	0.01	−0.13	0.03	0.02
	psychology	0.03	−0.05	0.06	0.03	0.08	0.04	0.05	−0.01	0.15	0.02

Table A.9.: Coefficients for baseline persona effects by dataset and category. Bold values indicate $p < .05$.

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Dataset	Category	Length			Audience Level	Stat. vs Dyn.
		Stat.	Dyn.	Com.		
MMLU-Pro	Overall	0.03	0.02	0.02	0.02	0.02
	biology	0.02	0.02	0.02	0.04	0.01
	business	0.07	0.01	0.04	0.04	−0.06
	chemistry	0.03	0.02	0.02	0.01	−0.02
	computer science	0.03	0.00	0.02	−0.01	−0.06
	economics	0.04	0.01	0.03	0.04	−0.06
	engineering	0.00	0.01	0.00	−0.04	0.03
	health	0.03	0.04	0.02	0.10	0.10
	history	0.02	0.03	0.02	0.03	0.08
	law	0.02	0.03	0.02	0.04	−0.02
	math	−0.01	0.00	−0.01	0.01	0.06
	other	0.05	0.03	0.04	−0.02	0.03
	philosophy	0.04	0.08	0.05	0.00	0.15
	physics	0.05	0.03	0.03	0.08	0.04
	psychology	0.01	0.00	0.01	0.02	0.04
Alpaca	Overall	−0.07	0.31	0.10	0.42	1.34
MATH	Overall	0.01	0.03	0.02	0.07	0.03
	Algebra	0.00	0.07	0.03	0.06	0.09
	Counting & Probability	0.01	0.01	0.01	−0.04	−0.06
	Geometry	0.03	0.01	0.02	0.10	0.01
	Intermediate Algebra	0.08	0.08	0.06	0.12	0.04
	Number Theory	0.05	−0.01	0.02	0.00	−0.10
	Prealgebra	−0.06	−0.03	−0.05	0.12	0.00
	Precalculus	0.03	0.08	0.06	0.10	0.16
IFBench	Overall	−0.05	−0.04	−0.05	−0.01	0.00
FLORES+	Overall	0.03	0.10	0.04	0.52	0.26
	Arabic	0.10	0.09	0.07	0.21	0.12
	Chinese	0.02	0.11	0.04	0.65	0.27
	German	−0.02	0.07	0.02	0.73	0.30
	Spanish	0.04	0.07	0.05	0.67	0.21
	Swahili	0.00	0.15	0.04	0.49	0.44

Table A.10.: Coefficients for length, audience level, and static vs. dynamic effects by dataset and category. Bold values indicate $p < .05$.

Dataset	Length						Audience Level		Stat. vs dyn.	
	Stat.		Dyn.		Comb.					
	<i>LR</i>	<i>p</i>	<i>LR</i>	<i>p</i>	<i>LR</i>	<i>p</i>	<i>LR</i>	<i>p</i>	<i>LR</i>	<i>p</i>
MMLU-Pro	55.9	0.00	56.7	0.00	65.4	0.00	38.0	0.00	87.6	0.00
Alpaca	137.8	0.00	139.6	0.00	62.0	0.00	142.2	0.00	746.2	0.00
MATH	12.0	0.09	8.1	0.32	14.2	0.04	13.2	0.06	11.0	0.13
IFBench	5.3	0.62	5.2	0.63	6.0	0.53	3.1	0.87	5.9	0.55
FLORES+	64.3	0.00	11.0	0.13	38.3	0.00	421.5	0.00	43.8	0.00

Table A.11.: Likelihood-ratio tests for model-dependent effects. Cells report likelihood-ratio statistics and p-values. Bold values indicate $p < .05$.

A.5. Statistical Model Specifications

All statistical tests were implemented in Python using `statsmodels`. For binary outcome variables, logistic regression models were fitted using `smf.logit`. For ordinal outcome variables, ordered logistic regression models were fitted using `OrderedModel` with a logit link function. In all models, `C(model)` denotes a categorical fixed effect for model identity.

A.5.1. Binary Logistic Regression Models

Persona baseline tests

```
1 model = smf.logit(
2     "score ~ C(persona, Treatment(reference='helpful')) + C(model)",
3     data=df)
```

Listing A.1: Binary logistic regression model for the persona baseline test

If the helpful-assistant baseline was not available, the base condition was used as the reference category instead.

Persona length

```
1 model = smf.logit("score ~ length + C(model)", data=df)
```

Listing A.2: Binary logistic regression model for testing persona length

Persona length by model

```
1 model = smf.logit("score ~ length * C(model)", data=df)
```

Listing A.3: Binary logistic regression model with model interaction for persona length

Audience level

```
1 model = smf.logit("score ~ level + C(model)", data=df)
```

Listing A.4: Binary logistic regression model for testing audience level

Audience level by model

```
1 model = smf.logit("score ~ level * C(model)", data=df)
```

Listing A.5: Binary logistic regression model with model interaction for persona length

Static vs. dynamic persona

```
1 model = smf.logit("score ~ is_dynamic + C(model)", data=df)
```

Listing A.6: Binary logistic regression model for testing persona type

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Static vs. dynamic persona by model

```
1 model = smf.logit("score ~ is_dynamic * C(model)", data=df)
```

Listing A.7: Binary logistic regression model with model interaction for persona type

A.5.2. Ordinal Logistic Regression Models

Persona baseline tests

```
1 model = OrderedModel.from_formula(  
2     "score ~ C(persona) + C(model)", data=df)
```

Listing A.8: Ordinal logistic regression model for the persona baseline test

Persona length

```
1 model = OrderedModel.from_formula(  
2     "score ~ length + C(model)", data=df)
```

Listing A.9: Ordinal logistic regression model for testing persona length

Persona length by model

```
1 model = OrderedModel.from_formula(  
2     "score ~ length * C(model)", data=df)
```

Listing A.10: Ordinal logistic regression model with model interaction for persona length

Audience level

```
1 model = OrderedModel.from_formula(  
2     "score ~ level + C(model)", data=df)
```

Listing A.11: Ordinal logistic regression model for testing audience level

Audience level by model

```
1 model = OrderedModel.from_formula(  
2     "score ~ level * C(model)", data=df)
```

Listing A.12: Ordinal logistic regression model with model interaction for audience level

Static vs. dynamic persona

```
1 model = OrderedModel.from_formula(  
2     "score ~ is_dynamic + C(model)", data=df)
```

Listing A.13: Ordinal logistic regression model for testing persona type

Static vs. dynamic persona by model

```

1 model = OrderedModel.from_formula(
2     "score ~ is_dynamic * C(model)", data=df)

```

Listing A.14: Ordinal logistic regression model with model interaction for persona type

A.5.3. Likelihood-Ratio Tests

```

1 def collect_likelihood_ratio_result(
2     df: pd.DataFrame,
3     dataset: str,
4     full_test_fn: Callable,
5     constrained_test_fn: Callable,
6     category_name: str | None,
7 ) -> list:
8     """Compare a full model against a constrained nested model.
9
10    The likelihood-ratio test evaluates whether the additional parameters
11    in the full model significantly improve model fit compared to the
12    constrained model.
13
14    Args:
15        df: Input data for the statistical test.
16        dataset: Name of the dataset or benchmark.
17        full_test_fn: Function fitting the full model.
18        constrained_test_fn: Function fitting the constrained model.
19        category_name: Optional subcategory name.
20
21    Returns:
22        A list containing the dataset name, category name,
23        likelihood-ratio statistic, degrees-of-freedom difference, and
24        p-value.
25    """
26    full_model = full_test_fn(df)
27    constrained_model = constrained_test_fn(df)
28
29    lr_statistic = 2 * (full_model.llf - constrained_model.llf)
30    df_difference = full_model.df_model - constrained_model.df_model
31    p_value = chi2.sf(lr_statistic, df_difference)
32
33    return [
34        dataset,
35        category_name,
36        lr_statistic,
37        df_difference,
38        p_value,
39    ]

```

Listing A.15: Likelihood-ratio test for comparing nested statistical models